New Scientist

WEEKLY 17 October 2026

HANNAH FRY ON THE AI CRISIS ENGULFING MATHS

A REVOLUTIONARY WAY TO BEAT EVEN THE WORST CANCERS

HOW EATING SUGAR AS A CHILD CHANGES YOU FOREVER

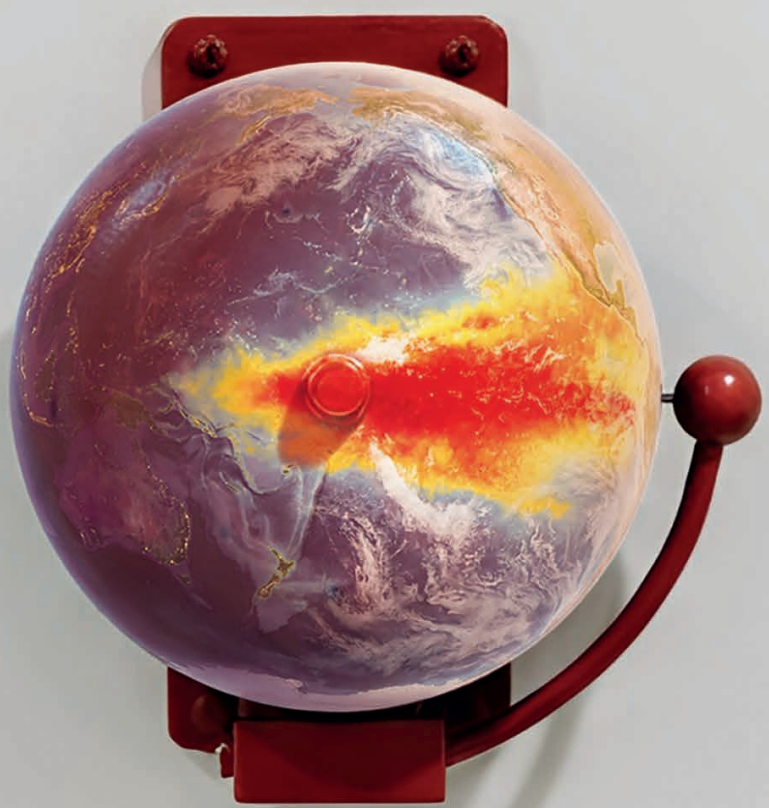
SPECIAL REPORT

EL NIÑO EMERGENCY

HOW IT BECAME A RECORD-BREAKING MONSTER

WHY IT COULD IRREVERSIBLY CHANGE OUR PLANET

THE MOST AMBITIOUS WAYS TO STOP IT



PLUS

ULTRA-FAST ASTEROIDS

THE NEANDERTHALS' WOODEN AGE

INVISIBLE TATTOOS

No3617 £7.95 CAN$11.99

9 770262 407398 42

Alzheimer's Society

Your Will has the power to make breakthroughs



Do something powerful by leaving a gift in your Will and help beat dementia

Dementia is overwhelming and heartbreaking – but it's not inevitable. Your Will has the power to help beat it.

We're the only charity tackling dementia from every angle by providing care for those who need it now, campaigning for dementia to be the priority it should be, and funding groundbreaking research led by scientists like Dr Ryan West.

This is only possible thanks to the kindness of individuals choosing to include gifts in their Wills, which together, fund a third of all this work.

We're determined to beat dementia, and with your support, that future can become a reality.

Your Will is more than a document; it reflects what matters to you and the future you want for your loved ones and generations to come.

We're at a turning point, and together we have the power to create a world free from the heartbreak of dementia.

Will you do something incredible and leave a gift in your Will to Alzheimer's Society?

Request your free guide to gifts in Wills

alzheimers.org.uk/willguide

0800 472 5264



☐ Please send me my free guide to leaving a gift in my Will

Title First name Last name

Address

Postcode

Email Phone

We'd love to send you updates on our life-changing services and research, as well as our fundraising, campaigning and volunteering activities.

Please tick below to let us know you're happy for us to stay in touch.

☐ **By email** ☐ **By phone** ☐ **Please don't contact me via post**

Alzheimer's Society is a registered charity in England and Wales (296645) and the Isle of Man (1128)

We will never sell your details to any third parties, please see **alzheimers.org.uk/privacy** for more information. If you'd like to change what you receive from us at any time, please call Supporter Care on **0330 333 0804**

Post in your own envelope to:
'FREEPOST ALZ SOC WILL GUIDE'
You do not need a stamp or any other address details.

Registered with FUNDRAISING REGULATOR

ALZLEG0034
LGPRAQA26G

This week's issue

On the cover

4 El Niño emergency
How it became a record-breaking monster

Why it could irreversibly change the planet

The most ambitious ways to stop it



Vol 272 No 3617
Cover image: Ryan Wills/Adobe Stock

19 Hannah Fry on the AI crisis engulfing maths

36 A revolutionary way to beat even the worst cancers

20 How eating sugar as a child changes you forever

15 Ultra-fast asteroids
15 The Neanderthals' wooden age
16 Invisible tattoos

36 Features

"Being able to identify exactly what makes a cancer tick is having an incredible impact"

News

14 Hidden health risk
Nearly 1 in 10 younger adults have plaque in their arteries

14 Giant finding
New species of giant waterlily described in Colombia

16 Ageing gracefully
How a 194-year-old tortoise's DNA fights off ageing

Views

19 Maths' existential crisis
Hannah Fry on what's happening with AI and maths

20 Sweet tooth
Alice Klein looks into how much sugar infants should eat

22 Aperture
A look back at all the fun of New Scientist Live

26 Culture
A compelling book argues we should cautiously embrace AI

31 Letters
Simulated reality question opens a can of worms



ALISTAIR YERYARD

22 A roarsome time New Scientist Live brought delight to attendees

Features

32 AI's inner life
Do AIs already process data in a way that gives rise to consciousness?

36 Cancer in the crosshairs
A new way of targeting tumours is helping people survive even the worst cancers

40 Panoramic cosmic views
NASA's latest telescope will help solve our biggest questions

The back pages

45 Puzzles
Try our crossword, puzzle and quick quiz

46 *Almost* the last word
Are humans the only animals that cry?

46 Tom Gauld for *New Scientist*
A cartoonist's take on the world

48 Feedback
A look into an unusual *Star Trek* partnership

48 Twisteddoodles for *New Scientist*
Picturing the lighter side of life



Time for that taste of Taylor's



Bracing for El Niño

We know how the climate will respond in the coming months, but are we ready?

EL NIÑO brings weather of opposite extremes – floods, droughts, cyclones, wildfires – to opposite sides of the planet. But we often forget that most of this extreme weather is highly predictable.

For example, all eight strong El Niños in the past 70 years have caused drought in Indonesia. All major models predict the current record El Niño will lead to a dry autumn and winter there. So it's little surprise that Indonesia is struggling with worsening drought and wildfires. Similarly predictable effects will be seen worldwide over the next year (see page 4).

Because of this predictability, some experts argue that there are no "natural" disasters. With enough preparation, rich societies should be able to withstand almost any extreme weather. Some governments, like those of California and Peru, have begun this preparation and declared states of emergency, but many haven't. China, India and some European nations have large grain stores to deal with food shocks (see page 8), but the US, the UK, Canada and Australia don't.

"With preparation, rich societies should be able to withstand almost any extreme weather"

Meanwhile, the most vulnerable countries – Ethiopia, Sudan and South Sudan – have plans to prevent widespread hunger and death, but other United Nations members have provided only one-third of the necessary funding.

Besides pledging money to those who need it most, investing more in the public infrastructure we use every day would pay off. UK sewers overflow and US and European power grids collapse even during "normal" conditions, so it is no wonder they aren't ready for extreme weather from El Niño or global warming. More speculatively, there are proposals to dull the worst impacts of El Niño through geoengineering efforts (see page 10).

Though El Niño is a natural cycle within Earth's delicate climate system, it's within our power to avoid its worst effects. That would mean, as ever, cutting carbon emissions to halt global warming, leaving us with one less extreme to worry about. The impact of ever-growing emissions is entirely predictable, so no one can say we weren't warned. ▮

PUBLISHING & COMMERCIAL

Display advertising
Tel +44 (0)203 615 6456
Email displayads@newscientist.com
Sales director Claudia Nicoletti
Partnership growth lead Joe Teal
Agency partner Daniel Sinclair

New Scientist Events
Tel +44 (0)203 615 6554 **Email** live@newscientist.com
Sales director Jacqui McCarron
Sales manager Maureen Ignacio
Head of product delivery Henry Gomm
Events and projects manager Georgia Hill
Events and projects executive Ciara Lucas

New Scientist Discovery Tours
Email tours@newscientist.com
Director Kevin Currie
Head of travel marketing Ginny Smith
Head of travel product Lara Paxton
Travel product manager Pip Orchard

Marketing & Data
Chief customer officer Claire Overstall
Head of campaign marketing James Nicholson
Subscription acquisitions manager Molly Hayde
Campaign marketing coordinator Charlotte Weeks
Head of customer experience Emma Robinson
Engagement marketing manager Kelly Spillane
Senior marketing executive Sarah Fabian
Head of CRM & audience data Rachael Dunderdale
Analytics manager Gary Parung
Data analyst Simonas Petridis
Junior data analyst Joseph Cole
CRM manager Ffion Evans
CRM executive Komal Nadeem

Technology
Director of engineering Michael Ilett
Principal engineer Phil John
Senior engineers Amardeep Sian, Maria Garrido, Matt Watson
QA engineer Kaveri Daitkar

Product
Head of product Remy Becher
Senior product manager Thomas Reader
Senior product designer Jacopo Verdini
Product manager Caitlin Jenkins
Head of product design Irmak Sunal Lutkin
Digital marketing designer Katarina Pollu
Brand marketing designer Laura Spanu
Senior product designer Sophie Majecki
Senior agile delivery manager Helen Garcia

NewScientist

Chief executive Roland Agambar
Chief operating officer Jonas Hermans
Chief financial officer Depak Patel
Executive assistant Lorraine Lodge

Finance & operations
Head of finance Charlotte Lion
HR business partner Tinka Bleijenberg

Partnerships
Consultant editor Justin Mullins
Editorial director B2B Edward Glanville
Programme editor Bailey Sellfors
Head of strategic partnerships David Allard

CONTACT US

newscientist.com/contact

General & media enquiries
UK 9 Derry Street, London, W8 5HY
Australia 58 Gipps Street, Collingwood, Victoria 3066
US 51 Astor Place, 4th Floor, New York, NY 10003

UK Newsstand
Marketforce UK Ltd
Email mfcommunications@futurenet.com

Syndication
Tribune Content Agency **Tel** +44 (0)20 7588 7588
Email tca-articlesales@tribpub.com

Subscriptions
View the latest offers
newscientist.com/subscription
Tel +44 (0)330 333 9470
Email subscriptions@newscientist.com
Post New Scientist, Rockwood House, Perrymount Road, Haywards Heath, West Sussex RH16 3DH

© 2026 New Scientist Ltd, England. New Scientist is published weekly by New Scientist Ltd. ISSN 0262 4079. New Scientist (Online) ISSN 2059 5387. Registered at the Post Office as a newspaper and printed in England by Warners Midlands Plc

recycle
When you have finished with this magazine please recycle it.

EDITORIAL

Editor Catherine de Lange
Executive editor Timothy Revell
Managing editor Penny Sarchet

News
News editor Jacob Aron
Assistant news editors
Alexandra Thompson, Sam Wong
Reporters (UK) Madeleine Cuff, Michael Le Page, Alec Luhn, Jo Marchant, Matthew Sparkes, Carissa Wong
(Aus) Alice Klein, James Woodford

Digital
Head of audience Matt Hambly
Podcast editor Rowan Hooper
Head of editorial video David Stock
Senior social media manager Isabel Baldwin

Features
Head of features Claudia Canavan
Deputy head of features Joshua Howgego
Editors Abigail Beall, Alison George, Olivia Goldhill, Jacklin Kwan, Thomas Lewton

Culture and Community
Head of books and culture Alison Flood
Senior culture editor Liz Else

Magazine
Magazine editor Eleanor Parsons
Assistant magazine editor Sophie Berdugo

Subeditors
Chief subeditor Kelsey Hayes
Bethan Ackerley, Tom Campbell
Trainee subeditor Lucy Wall

Design
Art editor Ryan Wills
Joe Hetzel, Phoebe Watts

Picture desk
Picture editor Tim Boddy
Assistant picture editor Jenny Quiggin

Production
Head of publishing operations Joanne Keogh

New Scientist US
Senior editor Chelsea Whyte
Features editor Grace Wade
Subeditor Alexis Wnuk
Audience editor Gerardo Bandera
Reporters Leah Crane, Karmela Padavic-Callaghan

Environment

El Niño heralds a tumultuous year

As the effects of the strongest El Niño on record start to be felt around the world, *New Scientist* takes a look at what this means for us all

THE world faces a difficult period ahead, as the El Niño weather phenomenon looks set to wreak havoc.

Already, India is grappling with the impacts of one of the driest monsoon seasons on record. Meanwhile, in Indonesia, schools have closed and authorities have advised residents to stay indoors to avoid the toxic smoke from huge wildfires supercharged by El Niño. Hundreds of thousands of people have faced respiratory illness, and carbon emissions are on track to equal Indonesia's catastrophic 2015 fires.

In this eight-page special report, we take a look at what this means for the world – from how the next few months could play out (see below) to its impact on food security (see page 8) and how it will end (see page 11).

We will also look at how the worst El Niño in history prepared us for this supercharged event (see page 6) and at some of the more ambitious ideas to mitigate its impacts (see page 10). But first, how did this El Niño grow so much bigger than usual, and what will its long-term effects on the planet be?



RYAN WILLS FOR NEW SCIENTIST; ADOBE STOCK

MIND-BLOWING. Unprecedented. Off the charts. Godzilla-level. Superlatives have been flying about the current El Niño climate pattern, which is already the strongest on record months before it is expected to peak – and unusually, they are coming from scientists, rather than the media. So, what has made this such an exceptional event? And for how long will we feel its effects?

To understand this El Niño, we have to understand La Niñas, the opposite phase of the climate pattern. During La Niñas, trade winds blow west across the equatorial Pacific, pushing surface waters warmed by the sun towards Australia and Indonesia. This piles up a layer of water known as the "warm pool". And it really does pile up: sea level in the western Pacific rises by as much as 20 centimetres.

"It gives us a glimpse of what is coming if we continue to warm the planet with carbon"

The bigger the warm pool gets, the more unstable it becomes. If the trade winds holding it in place weaken or reverse direction, the warm water spills back across the Pacific, making surface waters along the equator abnormally warm – an El Niño event. So the extent to which an El Niño raises global temperatures and causes extreme weather depends on how big and hot the warm pool gets, and on how much of it moves east across the Pacific.

By late 2025, the heat content of the western Pacific was much higher than normal – although it wasn't the highest on record. "Based on that alone, one might not have expected this event to become as strong as it has," says Michael McPhaden at the US National Oceanic and Atmospheric Administration.

The second key factor was a series of strong bursts of wind blowing east, pushing the extra-hot warm pool towards South America. This isn't just a surface phenomenon, says McPhaden. Each wind burst creates a downward bulge in the layer of warm water, known as a Kelvin wave, which travels across the

How will El Niño affect the globe?

From wildfires to landslides to coral bleaching, Alec Luhn takes a look at what happened in previous El Niños to forecast what will happen this time

OCTOBER
Typhoons in Asia

El Niño encourages the development of typhoons over the Pacific, but also shifts where they form eastward. This means that when they head towards Asia, they cover more distance and suck up more heat energy and moisture before landfall.

Nearly two dozen typhoons hit East Asia in the autumn during the three previous super El Niños. In October 2015, "super typhoon" Koppu destroyed thousands of buildings and killed at least 47 people in the Philippines. This came about two weeks after typhoon Mujigae set off tornadoes and landslides in China, killing at least 20, and super typhoon Dujuan killed at least two in Taiwan.

This year, typhoons have already blasted the Philippines and Japan, and more are likely to strike as sea-surface temperatures continue to rise.



XINHUA/SHUTTERSTOCK

Typhoons have already hit East Asia this autumn, and more are expected

Pacific over the following six to eight weeks.

Every Kelvin wave increases the thickness of the warm layer in the eastern Pacific. Put another way, the bottom of the warm layer, or thermocline, becomes deeper. "The thermocline is currently more than 80 metres deeper than normal in some areas," says McPhaden.

The deeper the thermocline, the harder it is for colder water to well up and reach the surface. So we now have a huge amount of extra warm water spreading across the surface of the Pacific and transferring huge quantities of heat to the atmosphere, with repercussions across the globe.

The question of whether global warming is making El Niños stronger is unresolved, says Michael Mann at the University of Pennsylvania. "It's honestly an open question from my stance."

The number of observed El Niño and La Niña events in modern times is too small to reveal any link. A recent study of corals in the Galapagos suggests there is a link, but Mann says there are crucial gaps in the coral record that mean no firm conclusion can be drawn.



AGUS KURNIAWAN/NURPHOTO VIA GETTY IMAGES

Wildfires in Indonesia have been intensified by the super El Niño

Climate models are little help either, he says. "The models [are] nearly equally split as to whether the amplitude of El Niño events is likely to increase or decrease with anthropogenic warming." What is clear is that warming amplifies the damage done by El Niños.

The current super El Niño will also lead to a big spike in global warming, says Mann. "It gives us a glimpse of what is coming down the pike if we continue to warm the planet with carbon pollution. Warmer oceans, more evaporation, bigger flooding events, hotter continents and worse droughts leading to more extensive and destructive wildfires."

Once this monster El Niño ends (see page 11), its effects will continue to be felt. "I expect that this super El Niño event will drive multiple climate-regime shifts like we've seen after past super El Niño events," says hurricane specialist Jeff Masters. "Regime shifts tend to cause considerable trouble – significant disruptions to ecosystems and to the humans that depend on them."

For instance, this El Niño is expected to lead to a devastating drought in the Amazon that could kill off some of the remaining areas of rainforest. This would release huge quantities of carbon dioxide and also make the whole region permanently drier, because much of the rainfall in rainforests comes from the water evaporating from trees.

"This super El Niño event will drive multiple climate-regime shifts like we've seen after past ones"

Some effects will be temporary. For instance, the Indian Ocean can become warmer after El Niños – on top of the warming due to global heating – leading to floods in some nearby regions. Some of the worst floods in the Yangtze river occurred in the summer of 1998, after the El Niño that ended in May.

The biggest concern is that the current El Niño could lead ›

NOVEMBER-DECEMBER
Flooding and drought in Africa

Northern and eastern Ethiopia have seen crop failures and South Sudan is on the brink of famine after one of the driest June-September rainy seasons on record, a fall linked to El Niño.

As the El Niño strengthens through the autumn and winter, it is expected to bring heavy rainfall and flash floods to southern Ethiopia, Kenya and Somalia. Flooding killed almost 3000 people in eastern Africa during the 1997 El Niño and 700 people during the 2023 one.

At the same time, rainfall is forecast to reach the lowest levels in 75 years in countries like Malawi, Zambia and Zimbabwe during these months. That threatens an even worse maize harvest failure than during the 2015-16 super El Niño, when tens of millions of people went hungry and hundreds of thousands of livestock died.

JANUARY-FEBRUARY
Flooding and landslides in the Americas

During the super El Niño in early 1998, massive rainfall turned a desert in Peru into a lake, washed away roads and bridges in Ecuador and buried infrastructure like a goldmine in Bolivia, killing hundreds of people. Flooding and disease killed more than 9000 people in Peru in the 1982-83 super El Niño.

At least 52 people have already been killed in Peru this year by flooding and landslides linked to the "coastal El Niño", a phenomenon where the cold waters off Peru warm up that often precedes the wider tropical Pacific warming of El Niño.

Heavy rains often extend to North America. California, where sea levels are now being raised by a giant swell of warm El Niño water known as a Kelvin wave, suffered hundreds of damaging landslides and mudslides in February 1998.

to changes that last decades or more. Last year, Jong-Seong Kug at Seoul National University in South Korea and his colleagues published a study showing that the combination of extreme El Niños and global warming can produce climate-regime shifts.

After the 2015 super El Niño, there was a sudden increase in sea-surface temperatures in the Gulf of Mexico that has persisted ever since. The El Niño wasn't the only factor involved, says Kug, but it contributed to this jump.

The even-stronger El Niño now under way is more likely to trigger such shifts. "We are experiencing a very strong El Niño event," says Kug. "Based on our study, such an extreme event could substantially increase the likelihood that various components of the Earth system are pushed toward, or even across, regime-shift thresholds."

In other words, in some parts of the world, the climate will never go back to how it was. "Our civilisation is adapted to the climate of the 20th century, and it is costly and disruptive to be forced to deal with sudden climate regime-shifts," says Masters. ▮

Michael Le Page

Look back

Lessons from the worst El Niño to hit the planet

Madeleine Cuff

TO READ of the great famine that swept the globe in the 1870s is to read of a hellscape worse than your imagination could conjure. In China, India, Brazil and beyond, after years of drought and failed harvests, millions died of starvation. People rioted. They ate diseased animals, tree roots, mud, pets and strips of leather in desperate bids to stay alive. By many contemporary accounts, cannibalism was rife. Children were sold for a meal. Corpses were bricked into their homes by family members to stop them being eaten by hungry neighbours. Bodies littered the roadsides.

50 million

The number of deaths globally due to the 1877-78 El Niño

Those who fled the famine took shelter in filthy camps and shanty towns, which were then struck by deadly outbreaks of cholera, smallpox and other diseases. In China, when winter came and there was no money for fuel, people burned their own houses. The desperate then crowded together in underground pits for warmth. Society all but fell apart. Once the poor had died in a region, the wealthy succumbed.

When the rains finally returned, there was barely anyone left to sow the crops. The total death toll across India, China and Brazil between 1876 and 1879 is estimated at 20 million to 30 million. Globally, there may have been 50 million deaths. As historian Mike Davis writes in the book *Late Victorian Holocausts*, this was a disaster of "truly planetary magnitude".

The root cause of this catastrophe? A supersized El Niño event in 1877-78, which released vast amounts of heat from the ocean into the atmosphere, triggering severe disruption to global weather patterns.

Of course, scientists in 1877 weren't aware of the El Niño-Southern Oscillation cycle. There was rudimentary meteorological data collection under way: sea-surface temperatures were collated, rain gauges were monitored, barometric pressure was observed. Scientists knew something strange was happening to global weather patterns, but the cause was far from clear.

In the immediate aftermath of the El Niño, many researchers were convinced sunspots were to blame. It wasn't until the 1920s that physicist Gilbert Walker first identified the seesaw of atmospheric pressure across the tropics, known as the Southern Oscillation.

Then, in 1969, meteorologist Jacob Bjerknes connected changing sea-surface temperatures in the Pacific to the Southern Oscillation and mapped out the global dynamics of an El Niño. That was the "big bang" for El Niño research, says Michael McPhaden at the US National Oceanic and Atmospheric Administration, ushering in the idea that with the right data and equipment, El Niños could be identified in real time – and even predicted in advance.

Since the 1990s, scientists have been able to forecast these events months in advance. That

MARCH-APRIL
Cholera and malaria outbreaks

Rising waters can overwhelm sewage systems, contaminating drinking water and spreading diseases like cholera and leptospirosis, a bacterial infection that can be transferred from animals to humans.

At the same time, standing water and warmer temperatures can help mosquitoes multiply and increase the transmission of malaria and dengue fever.

The massive El Niño rainfall across the top of South America in 1998 led to deadly outbreaks of cholera in Peru and other countries that spring.

Meanwhile, torrential rains of up to 100 times the average in eastern Africa sparked a deadly outbreak of malaria as well as Rift Valley fever, a virus originating in livestock that can also be transmitted by mosquitoes.

MAY
Widespread coral bleaching

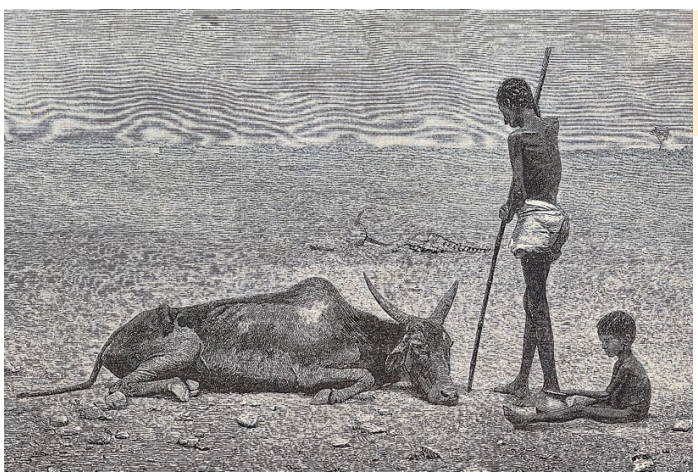


MAGEBROKER/MATHIEU FOULQUIE/ALAMY

Coral bleaching occurs as waters warm

Because El Niño hinders the upwelling of cold, deep water, all three previous super El Niños were accompanied by mass coral bleaching, when warming waters disrupt photosynthesis by symbiotic algae, and the coral host rejects the algae and essentially starves to death.

The US's National Oceanographic and Atmospheric Administration (NOAA) has said reefs across almost all of the planet's tropical oceans are now at high risk again. Sea-surface temperatures will probably peak at the end of austral summer.



DEA/BIBLIOTECA AMBROSIANA/GETTY IMAGES

The 1877-78 super El Niño led to a devastating famine in India

affords governments, emergency services, farmers and citizens vital time to prepare by increasing water storage, sowing drought-resistant crops, improving warning systems, restoring flood defences and more. "Being forewarned is being forearmed. If you know something is coming and it can have really damaging impacts, you have some choices to make," says McPhaden.

Models began to hint at a coming El Niño in February 2026, and by April there were clear signs that a strong event was looming. That gave governments a three-month head start before the El Niño even began, and at least six months of warning before impacts would be felt.

Improvements in El Niño forecasting, with the help of AI, mean it could soon be possible to predict a coming El Niño two years in advance, says McPhaden. "The longer you have to prepare, the more options you have. And the better off you may be in the end," he says.

But although there are prospects to improve El Niño forecasts even further, it comes at a time when US federal funding for climate research and global aid has been severely curtailed. In January 2025, President Donald Trump issued an executive order shutting down a US-funded famine early warning system. Six months later it was reinstated, but data reporting since then has been patchy, news reports suggest. The US has also severely cut funding to the World Food Programme, which provides emergency food relief. Other high-income countries, including the UK, have also scaled back their overseas aid budgets in recent years.

Meanwhile, the US war with Iran and blockade in the Strait of Hormuz have pushed up fuel and fertiliser prices, while Russia's war in Ukraine has curtailed the global wheat trade. This El Niño "is happening at a very bad time", says Richard Seager at Columbia University in New York (see page 8).

Back in 1877-78, the ineptitude of governments' official responses worsened the impacts. In India, colonial rulers continued to collect taxes, hoard grain and enforce food exports to England despite widespread starvation among local populations. In China, corrupt officials raided grain supplies, and poor transport links made it difficult to get food to those who needed it.

This time around, early preparation, coupled with a more comprehensive global trading network, better transport links and modern agricultural methods, means the world is much, much better prepared to deal with a super El Niño than it was 150 years ago. "We have much better public health systems," says McPhaden. "We have much better understanding of disease and disease controls. And we have humanitarian organisations, and we have social safety nets, that will not let 50 million people die like happened in 1877-78." ▮

"The longer you have to prepare, the more options you have. And the better off you may be in the end"

JUNE-JULY
Global heat

While El Niño is expected to peak at the end of 2026, it will take three to four months for all its heat to be transported around the globe. Because of this lag, and because it comes in addition to global warming, 2027 is expected to be the hottest year on record, reaching a maximum during the northern hemisphere summer.

Heatwaves can cause tens of thousands of excess deaths, while also reducing crop yields. During the 2023-24 El Niño, temperatures reached 4°C above average in a heatwave in South Sudan.



JENS SCHLUETER/AFP VIA GETTY IMAGES

Heatwaves can lead to water levels in rivers dropping

AUGUST
La Niña whiplash

NOAA projects that the Pacific will likely return to weak El Niño or neutral conditions in April, May or June. Forecasts don't extend beyond that yet, but strong El Niños in the past have often been followed by strong La Niñas, when enhanced winds increase the upwelling of cold, deep water. That cools the globe, but it also in many cases causes the opposite extreme weather, an especially devastating effect known as "climate whiplash".

Drought and wildfires can kill vegetation and create a crust over the soil, reducing the land's ability to absorb water. That increases flood and disease risk when La Niña finally brings rain. Some research has suggested that global warming will lead to wider swings between strong El Niño and strong La Niña, increasing the risk of whiplash in places like California and Spain.

Agriculture

A food crisis could be on its way

Droughts and flooding brought by El Niño threaten crops around the world, raising the risk of increased prices and famines, discovers **Alec Luhn**

EACH purple carrot the farmers pull from the clumpy soil of West Sussex bears the scars of extreme weather. Many are huge and bulbous, or even double-tipped. Record heat and widespread drought across the UK this summer killed some seeds and left the surviving ones more room to grow. Following the sudden return of rain this autumn, which was also abnormally warm, some carrots split down the side and tiny sap-sucking whiteflies cluster under their green leaves.

"If the crops are stressed, they're more likely to get attacked," says Tom Chadwick, who, along with his partner Rosa Farber, grows vegetables and fruit at Laines Organic Farm in the UK.

Rain pours down as Chadwick and Farber work, an omen of the warm, wet, stormy weather that El Niño is expected to bring to the UK this autumn and winter, which will stress the produce further. Too much rain could rot carrots still in the ground, as well as the garlic, salad greens and broad beans they will plant in the next few weeks.

But the farmers also fear that frost could kill their apples, plums and pears in late winter, when El Niño can disrupt the polar vortex and release Arctic air into the UK. Or that another dry summer could again require long days of emergency watering.

"It is worrying, the unpredictability of it," says Farber. "It's just an unknown."

Extreme weather from the super El Niño will hit every continent. Whether that is flooding and typhoons or drought and wildfires, it will damage crops and threaten a food crisis, experts say.

The result will be a tale of two socioeconomic statuses. People in high-income countries could face shortages and higher prices over the next 18 months. But those in low-income countries could suffer food insecurity or even famine.

"The UK may be willing to pay top dollar," says Christian Jaccarini at the Energy and Climate Intelligence Unit, a UK non-profit. "It doesn't mean people won't go hungry somewhere else."

"While countries like India have built grain stores to buffer these shocks, the UK and the US haven't"

After the hot, dry summer and its worst-ever cereals harvest, the UK now faces a "climate whiplash" to the opposite extreme. The last El Niño in 2023-24 led to the wettest six months ever observed in England and Wales.

Waterlogged soils can prevent tractors from getting onto the fields, or keep seeds from germinating. The harvest of grains like winter wheat and barley declined by more than 20 per cent in 2023-24, costing farmers £1 billion – and that was a far smaller El Niño than the current one. Prices of bread, biscuits and beer are likely to rise next year, according to the consultancy Parker Walsh. So are beef and dairy prices, as both abnormally wet and dry weather forces farmers to rely on expensive feed rather than grazing.

The worst impacts, however, are nearer to the tropics. Almost all recent El Niños have unleashed drought on Brazil, Vietnam, Colombia and Indonesia, the world's top four coffee producers, and commodity market prices of Arabica have already jumped 30 per cent. While consumers in Europe might have to cut back on coffee breaks, bean growers in South America and Asia could fall into poverty.

A recent UK government report warned chocolate prices could also increase. Dry weather is fuelling massive wildfires in Indonesia, the second-largest grower of cacao. Ecuador, the fourth-largest grower, is expected to suffer heavy rainfall and flooding.

But the most worrying threat is to rice, the primary staple for half the world's population. El Niño reduced summer monsoon rainfall in India, the biggest producer, by 13 per cent. The harvest in some parts of central India failed as farmers were never able to transplant seedlings from nurseries to the desiccated paddy fields. Yield is expected to fall by 7 per cent, or 10 million tonnes.

India is likely to restrict exports, according to Akshay Deoras at the University of Reading, UK, as it did during the 2023-24 El Niño, after which global prices shot up by 30 to 40 per cent.

"Inflation is a big challenge over the next few months," he says.

Multiple points of failure

Grocery chains in countries that import a large share of their food, including the UK, would typically try to buy rice from other providers. The problem is El Niño drought also threatens to wreck rice crops in Vietnam and Thailand, the next-largest exporters.

In addition, dry weather is also starting to decimate wheat crops in India and Australia, both major producers. California and Mexico, which supply most of the US's fruit and vegetables, face potentially damaging rainfall.

In the case of "multiple breadbasket failure", replacement supplies become difficult to find, and a staple can become brutally expensive or unavailable. While countries like China, Germany and India have built grain stores to buffer these shocks, the UK and the US haven't (see "The UK's food system is vulnerable", right).

"With El Niño, you have these phenomena of changes in weather patterns across the whole globe, which is part of why it makes it so difficult," says Monika Zurek at the University of Oxford. "You cannot easily substitute anymore."

The most vulnerable will be hit the hardest, since poorer families have to put a higher percentage of their income toward food, a



ALEC LUHN

El Niño is expected to bring storms to the UK this autumn, which will affect crop yields



FIRDOUS NAZIR/NURPHOTO VIA GETTY IMAGES

The most worrying threat is to rice harvests in countries like India

relationship known as Engel's law. Low-income Britons, who spend 70 per cent of their disposable income on meals, may be forced to buy lower-quality food. In lower-income countries, people can face food insecurity, which is defined as having access to fewer than 2500 calories a day.

West Africa is at risk of hunger due to its reliance on imported rice, according to the Famine Early Warning Systems Network. But the highest concern is southern and East Africa, where El Niño drought will be a "compounding shock" on top of existing problems like conflict. It could undermine

"If you have increases in food prices, then the whole backlash is going to be against the government"

sorghum and millet crops in Ethiopia, Sudan and South Sudan, which are already suffering some of the driest weather on record.

In south-eastern African countries such as Zimbabwe and Mozambique, the crucial maize crop could wither. During the 2015-16 super El Niño, maize fields became piles of dry husks, cattle pastures were littered with bony carcasses and 36 million people in Africa faced hunger. It could be even worse this time, as foreign aid decreased by 23 per cent in 2025 and the US Agency for International Development was abolished.

El Niño arrives as US-Iran tensions continue to choke diesel and fertiliser shipments through the Strait of Hormuz. A report by the think tank Safe Climate Australia argued that, in Asia, this combination could lead to "social unrest and conflict in vulnerable countries, much as the global wheat crisis became a trigger of the Arab Spring". Deoras also thinks it is a risk.

"If you have got increases in food and fuel prices, then the whole backlash is going to be against the local government," he says.

All of this comes as humanity approaches what is almost certain to be the hottest year in its history. Though it is far too early for forecasts, many farmers and experts fear another sweltering, dry summer in 2027. Laines Organic Farm is installing tanks to collect rainwater and hopes for government funding to drill a well.

"That's what all the farmers are talking about after that hot summer. Everyone should have their own borehole," says Farber. "The summer really wiped us out." ▮

Interview

"The UK's food system is vulnerable"

THE record El Niño climate phase will unleash unprecedented heat and extreme weather, and the UK food system is vulnerable, UK climate minister Katie White told *New Scientist* last month.

At the same time, she pushed back against "panic and fear" after environment minister Angela Eagle told Britons they should "store a bit of food that can keep you going for a while" in case of El Niño storms and flooding.

"Look, am I going out and buying loads of extra beans? I've always got a few cans of beans, but that's mainly so I'm ready for when I've not had a chance to go to the shop," said White. "So I'm not going out and doing that."

The UK is likely to face an unusually warm, wet and stormy autumn and winter due to El Niño (see main story, left). "We will have a wetter start to this winter... On the back of a very dry summer, that is difficult," said White. "The food system is vulnerable."

Large countries including China, Brazil and Germany have been building up stores of key foods like wheat, rice, corn and sugar. But the UK doesn't keep substantial public food reserves.

White argued that the UK would be better off helping local farms and groceries to become more resilient to climate shocks. A task force she established that includes military experts and scientists will meet this month to scrutinise readiness for climate security threats, including to food.

"I don't think we need to be in a position of grain stores, but I do think we do need to be looking at where the vulnerabilities are in the food system and how we support local production in the most sustainable way," she said. "That's going to be working with farmers and also on local supply chains."

Farmers will probably have to plant more weather-resistant varieties or different crops entirely, according to Monika Zurek at the University of Oxford. The UK also needs to reduce its reliance on imports for 40 per cent of its food. But "grain storage is definitely part of the answer" as well, she says.

El Niño flooding could also threaten housing and infrastructure, and these risks will continue as climate change makes most of the world warmer and wetter. The UK should not only build anti-flood engineering works such as the Thames Barrier, but also restore ecosystems like wetlands to absorb excess water, said White.

The long-term solution is for the UK and other countries to reach net-zero greenhouse gas emissions by 2050, she said.
AL

UK climate minister Katie White pushed back against "panic and fear"



DAVID DEE DELGADO/GETTY IMAGES FOR NYT

Technology

How to save the world from El Niño

Ambitious ideas to shield us from the worst impacts of El Niño are being worked on, but how feasible are they, asks **Madeleine Cuff**

AS COUNTRIES around the world brace for a torrid few months as El Niño hits, what if we could take the sting out of its tail – or even stop it from happening altogether? That's the hope of physicists, chemists and engineers working on ideas ranging from the outrageously radical to the just-might-work.

"Artificial upwelling could be used to potentially prevent an El Niño from forming"

One of the most ambitious ideas for halting El Niño is to repurpose a technique called artificial upwelling, which was initially conceived to boost the uptake of carbon by the oceans. This form of geoengineering involves pumping cold, nutrient-rich water from the deep ocean up to the surface to encourage the growth of plankton, which would then absorb and store carbon from the atmosphere.

Katharine Ricke at the University of California, San Diego, says artificial upwelling could be used to cool sea-surface temperatures in the tropical Pacific, potentially preventing an El Niño from forming.

But – and it is a big but – the technical feasibility of deploying artificial upwelling on this scale would be a "major, major undertaking", she says. Ricke has previously investigated whether the approach could be used to cool sea-surface temperatures in the Indian Ocean and found it would require approximately 5.8 million kilometres of pipe, nearly double the total length of all pipelines in the world.

There is also the problem that nobody knows what putting that amount of pipework into the ocean would do to other marine life. "The ecosystem effects of installing ocean pipes at that scale are really not well understood at all," says Ricke.

Another equally ambitious, but perhaps more promising, approach for preventing El Niño could be to use another geoengineering technique called marine cloud brightening.

The El Niño climate phase occurs when easterly winds weaken, allowing warm water built up in the western Pacific to slosh back across the central and eastern parts of the ocean. Spraying droplets of seawater into the air below low-lying stratocumulus clouds in the Pacific would increase their brightness, reflecting more sunlight back to space. This would bring down sea-surface temperatures, strengthening trade winds to blow warm water back into the western Pacific and reverse the El Niño pattern.

A study published in July, co-authored by Ricke, concluded that deploying marine cloud brightening in this way would dramatically reduce the strength of an El Niño and push the system back into a neutral state much earlier than normal.

This won't come in time to prevent the current El Niño, says Jessica Wan at the University of Chicago, a co-author of the research. The technology needed to brighten clouds on this scale doesn't exist, and there is no international framework for countries to agree on this kind of geoengineering intervention. "It's not something we could implement for this El Niño or maybe even the next one," she says.

Heat tolerance

So, for now at least, El Niño cycles aren't going anywhere – but can we make them more tolerable?

One particularly devastating consequence of the way El Niño turns up the global temperature dial and shifts rain patterns around the world is that it threatens food supplies (see page 8). The United Nations World Food Programme estimates that 274 million people in 45 countries will be acutely food insecure by the end of 2027, 49 million more than today.

Work is under way to develop new breeds of crops capable of withstanding the extreme conditions brought about with El Niño. Rice is a particular priority, says Erik Murchie at the University of Nottingham in the UK.

Murchie is part of a team developing heat-tolerant rice varieties, with trials now ongoing in the Philippines. This is part of a global effort: in 2025, researchers in China reported the discovery of a gene variant that can preserve yield and rice quality during hot nights, one of the major threats to rice yields. Gene editing or selective breeding could allow this trait in commercial crops, the team says, with the field trials indicating a 31 to 78 per cent greater yield under high temperatures compared with standard rice strains.



MARIO TAMA/GETTY IMAGES

Flood defences, like this one in California, are just one way to protect communities from El Niño

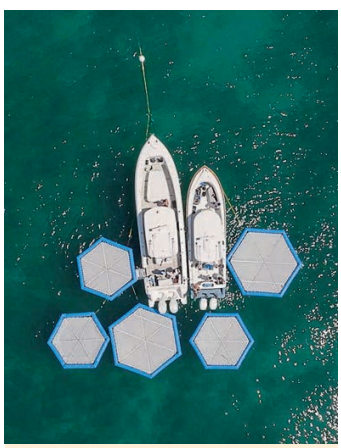
Such a goal isn't a fantasy – in the 1990s, scientists pinpointed a rice gene that allows a traditional rice variety known as FR13A to survive a week of submergence in flood water. The *SUB1* gene has since been introduced into several major rice varieties around the world, making crops more resilient to climate-driven flood risks.

Housing, people and infrastructure are also vulnerable to rising water levels and flooding driven by El Niño. Traditional flood-protection measures are already being deployed, from coastal barricades to temporary barriers. In California, drains are being cleared and sandbags handed out, while residents are being urged to prepare their properties for high winds and flood risk.

But one company has a more drastic idea to protect against flood risk more permanently: raising the height of land. California start-up Terranova has developed a robot that injects a mix of wood slurry into the ground, raising the height of land at risk of flooding from sea-level rise.

The robots drill wells around 12 to 18 metres deep into the ground and then inject the mix, made from wood waste, into the earth. The robotic system can lift up a roughly 4000-square-metre area by about 30 centimetres per day, the firm says. "We're combining heavy robotics and geotechnical innovation to literally reshape the world," the firm's CEO Laurence Allen said in a 2025 statement announcing $7 million of venture capital investment.

However, Nigel Wright at the University of Birmingham in the UK says extensive testing should be done before deploying the technology at a large scale. He queries whether the wood waste would solidify, remain as slurry or decompose underground over the long term. "I would question whether the soil structure would be stable enough over 20 or 30 years to actually support houses," he says.



UNIVERSITY OF MIAMI

Shading corals from the sun could help prevent coral bleaching

Inside the oceans, coral reefs are on the front line of the El Niño crisis. Elevated ocean temperatures caused by El Niño can cause mass coral-bleaching events, in which corals expel the symbiotic algae living in their tissues. This process can severely stress, and ultimately kill, corals.

> "Gene editing or selective breeding could add a heat-tolerant trait to commercial rice crops"

Shading corals could help. One test of the approach is happening this month: five floating hexagons that block harmful ultraviolet rays are being installed in a small coral nursery in Florida. The plan is to extend this to 100 shading units in June next year, which will be distributed among restoration organisations working across Florida to test their effectiveness at protecting corals and any impact on the long-term health of the ecosystem.

Coral sunscreen

In Australia, an approach that goes even further has been tested. Greg Qiao at the University of Melbourne set out more than a decade ago to create a chemical film that would sit atop agricultural reservoirs to minimise water loss to evaporation. After extensive trials, that idea never came to fruition – but Qiao realised the work could be repurposed for coral reefs.

The aim for a coral reef is to block sunlight, rather than prevent evaporation, so Qiao's team developed a thin film layer made from calcium carbonate coated with stearic acid. The layer is one molecule thick and designed to sit on the surface of the ocean above a reef. Field trials in 2018 suggested the film reduced light hitting the corals by 30 per cent. "That's probably enough for us to stop the bleach," says Qiao. The film would maybe last for only a few hours before dissipating, but that might be enough to offer protection for corals during a period of peak bleaching risk, he says.

Just like with many of the protection efforts against El Niño, coral shading is still in its infancy. There are promising signs that we may one day have a future without such dramatic global swings, but, for now, El Niño is here to stay. ▮

Weather

El Niño will come to an end – but when is harder to predict

Michael Le Page

WHILE the super El Niño will peak around November or December, when and how will it end?

One reason why El Niños end is that the warm water that spreads across the Pacific disperses. Some flows away from the equator, and the water also cools as it transfers heat to the atmosphere, which leads to a sudden jump in global surface temperatures (see page 7).

What happens below the surface also plays a key role. For instance, the arrival of warm water in the eastern Pacific triggers huge underwater waves known as Rossby waves that travel westwards. When these hit the continental shelf in the western Pacific months later, they bounce back eastwards as cold, faster-moving Kelvin waves. These thin the layer of warm water in the eastern Pacific when they arrive, helping deeper, colder water reach the surface and bringing the El Niño to an end.

The combined forecasts of computer models suggest that this El Niño will be over by June, but individual models vary. Some indicate it could persist well into the northern hemisphere's summer.

We can't predict with any certainty how this event will end, says Michael McPhaden at the US National Oceanic and Atmospheric Administration. The 1997 El Niño ended with a bang, with sea-surface temperatures in the eastern equatorial Pacific plummeting by 8°C in a month, followed by a strong, long-lasting La Niña. But the next strong El Niño in 2015 faded gradually and was followed by a weak, short La Niña. "So, at this point, it is difficult to know precisely how the current El Niño will wind down," he says. ▮

Analysis Artificial intelligence

OpenAI's shock dump

Artificial intelligence has been making impressive progress in mathematics. **Leah Crane** and **Matthew Sparkes** look at what OpenAI's recent mass paper release means for the field

THE floodgates of AI mathematics have opened. On 7 October, OpenAI released 722 mathematical papers using an AI model to examine an astonishing range of topics, from longstanding mathematical mysteries to small improvements in rarely considered curiosities.

Last month, OpenAI solved a problem related to the Navier-Stokes equations of fluid dynamics, which was one of the most enduring puzzles in mathematics. Having achieved impressive depth, OpenAI has seemingly now opted for breadth.

Francis Johnson at University College London worked for many years on Wall's D(2) problem – one of the hundreds of puzzles solved in OpenAI's tranche of papers.

"I worked on this problem for 25 years. I produced two books on it. I, personally, gave up," says Johnson. "I'm surprised that [AI] has done it quite so quickly, but I'm not surprised that it's done it."

He recounts working with AI models in recent months and being "astonished with the sophistication and clarity of the analysis that they gave".



AP PHOTO/JEFF CHIU/ALAMY

An unknown OpenAI model made the new mathematical discoveries

"Let's face it, the genie is out of the bottle now. We're going to have to live with it"

"Let's face it, the genie is out of the bottle now. We're going to have to live with it. Human beings are supposed to be adaptable, so we're going to have to adapt. I think for the moment, we just stand back and be astonished," says Johnson.

Of course, there hasn't yet been time for mathematicians to comb through all of these results, let alone to actually check them: some of the papers are hundreds of pages long, dense with complex computations. Many of them have not been formalised – reduced to steps that can be checked by a computer – meaning there is little assurance that the calculations are accurate. As *New Scientist* went to press, three had already been retracted. Nevertheless, even a preliminary comb through the thicket of papers turns up some that are likely to make a splash.

Several relate to Millennium Prize Problems, a set of seven mathematical problems generally considered to be among the most important and challenging in the field. Solving any of these problems comes with a $1 million prize from the Clay Mathematics Institute. Since its inception in 2000, one has been solved by a human mathematician, and one – the Navier-Stokes problem – seemingly by AI.

Now, OpenAI has taken aim at the Riemann hypothesis, an unproven rule about the distribution of prime numbers that, if proven, would map how prime numbers are spread out across the number line. OpenAI isn't claiming to have proven the Riemann hypothesis itself, but a variation called the quasi-Riemann hypothesis that deals

AI finds largest symmetrical Venn diagram yet

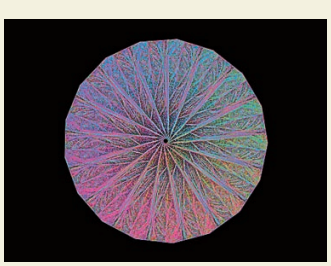
While AI is making many impressive mathematical discoveries (see main story), the most aesthetically pleasing may be a lace-like Venn diagram with tens of thousands of segments and rotational symmetry.

Venn diagrams show all possible relationships between sets, and are commonly used and understood with three or four circles.

Mathematicians have long been fascinated by the idea of hunting for symmetrical Venn diagrams of increasing size. It has been proven that they can only be created with a prime number of sets, but when these are kept "simple" – meaning no more than two curves cross at any point – they become even harder to find.

In 2012, an 11-set Venn diagram that met those criteria was discovered, and another with 13 sets was found in 2014.

Software developer Chris Dzoba decided to try using AI to find the next largest diagram, one with 17 sets. He set up a message board on which two AI models, Anthropic's Claude Fable and OpenAI's GPT-6 Astra, could communicate, then asked them to work together to find an algorithm that would provide a solution.



CHRIS DZOBA

Two AI models made a beautiful symmetrical 17-set Venn diagram

He started work on a Tuesday and by Thursday the models had unearthed a 17-set diagram. Next, he set the models to work finding a 19-set diagram and landed on a solution that Sunday (arXiv, doi.org/rr86).

He is now searching for 23 sets – although that will require paying for cloud computing to run the algorithm. MS

with how far prime numbers can be from their expected position. If the result stands up, it could be the biggest step towards a solution to the original hypothesis in years.

Two other Millennium Prize Problems are in the papers: the Birch and Swinnerton-Dyer conjecture and the Hodge conjecture. Like for the Riemann hypothesis, the papers on these problems claim to prove simpler, partial versions of the conjectures that could help lead to full solutions.

The full spectrum

Other papers provide minuscule improvements in algorithmic power. One claims that multiplication of integers could, in theory, be performed 2^{-182} faster than researchers previously thought: if a multiplication currently takes a millisecond, then the improvement is a little less than 1 divided by 14 million billion billion billion times the age of the universe. In other words, it is a very small improvement. Still, the papers show that improvements are possible.

Not all of the problems tackled in the OpenAI papers are quite so technical. For example, one is about the problem of colouring individual points on a plane so that no two adjacent points are the same colour. We already knew that this would require between five and seven different colours, and this new work seems to eliminate the possibility that it is five.

The bottom line is that with such a huge body of research all dropping at once, it could take years for mathematicians to check, and far longer to understand its significance, if it holds up to scrutiny. If AI maths continues at this pace, it could be a real boon or a real problem for the field of mathematics as a whole, and there is no way to predict which it will be until the work is validated.

In fact, OpenAI has recently faced criticism over the way it has disclosed mathematical research. Terence Tao at the University of California, Los Angeles, said last month that how discoveries are being made and "dumped" on the mathematical community may harm the field. In response, an independent Advisory Group on Mathematics and Artificial Intelligence was set up to advise AI companies on best practice.

OpenAI has previously said that it would take this advice on board, but its most recent release seems to go against many of the group's suggestions. For instance, it recommends that the model's name, the prompts used and the estimated cost of computation be released. OpenAI, however, didn't name the AI that made the discoveries, saying only that it was an "internal frontier model".

Lindsay McCallum Rémy at OpenAI told *New Scientist* in an email announcing the results that "we're continuing to explore other community-hosted alternatives for this release which meet the committee's guidelines. For future releases, we are committed to further improving the quality of the papers via the citations, mathematical exposition, and presentation of the results for better understanding. We want to give the mathematical community time and space to assess this work." ▮

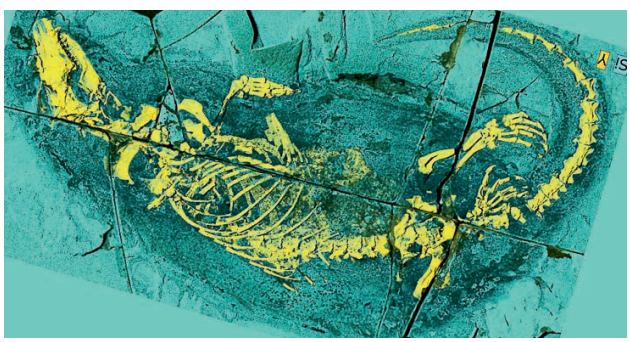
"With such a huge body of research all dropping at once, it could take years for mathematicians to check"

Read Hannah Fry's take on OpenAI's recent announcements on page 19

Palaeontology

Fossil mammal is earliest evidence of suckling

James Woodford



LUO ET AL., NATURE

M. sungi is preserved with fossilised bones and soft tissue (dark blue)

A SEMI-AQUATIC forebear of all living mammals that lived between 163 million and 168 million years ago was the first known animal capable of feeding by suckling.

Zhe-Xi Luo at the University of Chicago and his colleagues have discovered a new genus of mammaliform, a creature that lived before the last common ancestor of the living mammalian groups – placentals, marsupials and monotremes – that would have been able to chew and suck.

Found in the Daohugou area of Inner Mongolia, China, it is also the earliest known example of a semi-aquatic mammal. The animal has been dubbed *Megacauda sungei*.

"The back half of *Megacauda* is more similar to the platypus, of all mammals. But the skull and teeth are analogous to river otters," says Luo.

With an estimated weight of 1 kilogram, the creature is the largest known mammaliform from the Middle-Late Jurassic, when the planet was dominated by dinosaurs.

The pelvis is preserved in 3D, as are its hyoids, or throat bones. The bone structures under the skull that form the roof of the mouth and throat passage are also visible. This meant that, for the first time in any mammaliform, the team were able to reconstruct the animal's entire throat anatomy (*Nature*, doi.org/rr85).

"The mammalian ways of swallowing and suckling are unique," says Luo. Mammals' throats can form "a suction vacuum for milk and squeeze the chewed food balls and liquid down their throats", he says. All the features that enable this are present in *M. sungei*.

"We used to know very little about the evolutionary pattern of pharyngeal and hyoid structure. Now we know their evolutionary origins went back to before the rise of the modern mammals," says Luo.

Oliver Griffith at Macquarie University in Sydney says this fossil sits at a key point in mammalian evolution. "It tells us that some key mammalian traits like sucking, which is required for behaviours like drinking milk from mothers, predate the origin of mammals today."

He says it is "cool to see that this sister lineage to today's mammals had evolved an aquatic lifestyle, which is something we have seen multiple times across the mammalian family tree". ▮

Health

Fatty arteries present in young adults

Atherosclerosis, the build-up of fatty plaque in arteries, is "not a disease of old age"

Christa Lesté-Lasserre

NEARLY 1 in 10 younger adults have plaque in their arteries, according to a study of almost 17,000 people. Known as atherosclerosis, the build-up of fatty plaque can cause narrowing that reduces blood flow, leading to heart attacks and strokes. Atherosclerosis is more common in people aged over 50, so younger individuals may be assumed to be at low risk.

"The most important message is that atherosclerosis is already present in a meaningful fraction of people in their 20s and 30s," says Olli Raitakari at the University of Turku in Finland, who wasn't involved in the study. "Atherosclerosis is a lifelong process, not a disease of old age."

Previous studies have already suggested that atherosclerosis can occur surprisingly early. In 2025, Raitakari and his team found plaque in the carotid arteries – which supply blood to the head, face and neck – of about 5 per cent of Finnish adults aged under 30. But there hadn't been large-scale investigations spanning multiple adult age categories that examined several artery locations.

To fill this gap, Borja Ibáñez at the Spanish National Centre for Cardiovascular Research in Madrid and his colleagues recruited almost 17,000 adults, aged 18 to 70, in Denmark and Spain. They divided the participants, who had no known atherosclerotic cardiovascular disease, into five age groups, starting at 18 to 29. The study was made up of around the same number of men and women.

"Atherosclerosis is already present in a meaningful fraction of people in their 20s and 30s"

The researchers looked for plaque using CT angiography, which creates detailed 3D images of blood vessels, for arteries supplying the heart. They also used ultrasound for arteries in the neck and legs. Overall, they found "silent" atherosclerosis in 57 per cent of the participants, including 63 per cent of men and 51 per cent of women. This is defined as plaque that isn't causing any symptoms, but may lead to cardiovascular issues down the line.

In the youngest group, 9 per cent of the men and 7 per cent of the women had some degree of atherosclerosis. Among those in their 30s, 35 per cent of the men and 21 per cent of the women had atherosclerosis. The numbers continued to rise with each decade, with women beginning to catch up with men around the age of menopause.

Atherosclerosis was generally more common among the participants with known risk factors, such as higher LDL cholesterol and blood pressure or a family history of the condition. But these factors didn't reliably predict the presence of plaque, particularly among the younger adults (*The New England Journal of Medicine*, doi.org/hcg643).

The findings suggest that measures to detect plaque could identify people who seem to be at low risk of atherosclerosis, but might benefit from earlier preventive measures, such as medication or dietary changes.

But the presence of plaque doesn't necessarily predict a future heart attack or stroke, says Raitakari, and there is no evidence that screening people without known health problems would prevent such events. The researchers also state that it is unclear if the results apply outside of Denmark and Spain. ▮

Botany

New species of giant waterlily described in Colombia

GIANT waterlilies aren't exactly inconspicuous plants. But a new species has just been described, identified with the help of social media and a jungle dash.

Giant waterlilies are famed for their massive leaves, which grow up to 3 metres wide. For many decades, just two species were known: *Victoria amazonica* and *Victoria cruziana*. Then in 2022, *Victoria boliviana*, was added to the list.

In 2021, Jeremiah Harris, who runs a carnivorous plant business in the US, posted pictures online that he took while swimming near the Inírida river in Colombia next to what he assumed was *V. amazonica*.



CARLOS MAGDALENA; MAURICIO BERNAL



Carlos Magdalena was part of the team that named the giant waterlily *Victoria colombiana*

Carlos Magdalena at the Royal Botanic Gardens in Kew, London, was struck by the photos because scientists had thought there were no giant waterlilies in this region.

Magdalena and his team examined satellite photos and found telltale green dots confirming there really were giant waterlilies growing there. They then organised an expedition to the region.

The team found two plants but neither was flowering – and analysing the flowers is a crucial step in establishing whether a waterlily really is a new species.

On the last night, one of the guides, Gustavo Melo from the Indigenous Tucano community, said he knew where some more giant waterlilies were growing. The team set off by boat in the dark, and found five plants in a small lake, including some that were flowering.

The team has now published a paper describing the new species, which they have named *Victoria colombiana* (*Phytotaxa*, doi.org/hcmxf2). It differs from *V. amazonica* in the orientation of the spines beneath the leaves, and in the shape of its flowers and fruits. ▮

Michael Le Page

Space

Two asteroids are spinning faster than we thought possible

Jonathan O'Callaghan

LARGE asteroids have been found spinning so fast they should break apart, suggesting they might be stronger than we thought.

Dmitrii Vavilov at the University of Washington in Seattle and his colleagues analysed some of the first data collected by the Vera C. Rubin Observatory in Chile, which began a 10-year survey of the sky in June 2026. From a sample of more than 9000 asteroids, they measured the rotation of 260, based on a change in their brightness caused by the spinning asteroids' irregular shapes.

Of these, 66 were deemed to be super-fast rotators, spinning quickly enough to complete a rotation in less than 2.2 hours, a limit at which asteroids should no longer be able to hold themselves together under their own gravity.

Two asteroids in the belt between Mars and Jupiter were even more unusual ultra-fast rotators. Each was larger than 2 kilometres wide, but completed a spin every 5 minutes, a speed never seen before for such big asteroids (arXiv, doi.org/rsc2).

"On the scale of billions of years, asteroids of this size should hit each other and reaccumulate," says Vavilov, forming loose collections of material known as rubble piles that would break apart when rotating rapidly. The high rotation speed suggests that these asteroids aren't rubble piles, but instead monolithic objects at least 30 times stronger than clay, says Vavilov.

This probably means our models of how asteroids have smashed together over the 4.6-billion-year history of the solar system aren't correct, he says. That might mean these are pristine objects from the early solar system.

The asteroids might have been spun up by the sun heating their surfaces and giving them torque, known as the YORP effect, but that in itself is uncertain because of their large distance from the sun. ▮

Ancient humans

Neanderthal Europe was as much a wooden age as a stone age

James Woodford



GERARD CAMPENY/IPHES-CERCA

The imprint of this wooden club-like tool is unlike anything else seen

WOODEN tools, weapons, building materials and domestic items discovered in a Spanish cave challenge the assumption that Neanderthals' technological life was based primarily on stone.

The finds have given an unprecedented glimpse into everyday Neanderthal life and are a rare opportunity to recover part of their missing world, says Josep Vallverdú at the Catalan Institute of Human Paleoecology and Social Evolution in Spain.

"Neanderthal technology was not only made of stone; it was also made of wood," he says. "The archaeological record gives us a very biased picture of prehistoric technology. We see the stone because it survives, while a huge part of their material culture has simply disappeared."

Vallverdú and his colleagues have now completed the first systematic review of nearly 600 wooden objects, all between 40,000 and 62,000 years old.

The finds have been collected by generations of archaeologists since the 1980s, excavated at a rock shelter site called Abric Romaní near Barcelona. They include timber imprints in travertine – a rock rich in calcium carbonate that precipitates out of water in certain environments. The rock preserves the patterns of the wood grain even though the timber long ago rotted away. Charred artefacts and actual preserved timber have also been found at the site.

The researchers' first job was to determine exactly which pieces were used or made by Neanderthals. Only 13 qualified.

María Gema Chacón, also at the Catalan Institute, says the researchers have been surprised at how "complex Neanderthal woodworking could be, and how well they understood the properties of wood", with two spears being especially striking to her.

"The technological behaviours preserved are comparable to those seen for *Homo sapiens*"

"Producing long, straight shafts and shaping them deliberately into pointed weapons required planning, skill and careful selection of the raw material," says Chacón.

The technological behaviours preserved at Abric Romaní are comparable to those documented for *Homo sapiens*, she says.

Other finds included a mysterious wooden tripod, as well as timber trays reminiscent, in terms of likely utility, of the coolamons traditionally used by Indigenous Australians to carry food and water. Some of the wooden objects appear to be structural poles.

But perhaps the item that has most surprised the team is a "club-like" object that has no precedent from any other palaeolithic site (*Science Advances*, doi.org/hcmw9d).

Vallverdú says the site shows that wood was a normal and important part of Neanderthals' everyday life.

"We have evidence of woodworking across many archaeological levels and over thousands of years, and the objects are associated with very different activities: hunting, domestic tasks, containers and the organisation of living spaces," he says.

Chris Stringer at the Natural History Museum in London says the finds reveal a much wider technological repertoire of woodworking skills.

"This is of course evidence from only one site, in one area, and one time span," says Stringer, "so it's just the tip of the iceberg in terms of a whole and mostly hidden technology that ran in parallel with the much better-known stone tools of the Palaeolithic." ▮

Technology

Light could turn tattoos on and off

James Woodford



MATTER & LIGHT, BRUNS ET AL. 2026

"PhotoTat" tattoos (left) disappear when exposed to strong visible light (right)

TATTOOS may be radically reimagined following the development of inks that can appear or disappear depending on light exposure.

"You can simply turn the tattoo off whenever you don't want to see it and it disappears from view," says Carson Bruns at the University of Colorado Boulder.

Bruns has had a fish scale tattoo using this method for the past two years, and says he has had no known side effects and no fading.

Instead of traditional inks, Bruns and his colleagues employ photochromic dyes, which change from a transparent state to a coloured one when exposed to certain wavelengths of light. They call the new technology, which has been tested on silicone, "PhotoTat".

There is little mechanical difference between a traditional tattoo and the new inks, says Bruns. Both are pigment particles implanted in the dermis with a needle, he says. They also both stay there permanently and are applied with the same methods.

The difference is the pigment, says Bruns. With PhotoTat, "ultraviolet light writes the colour in, bright visible light erases it, and the cycle can be repeated", he says.

"The tattoo is permanent, but its appearance is not."

This could be useful in medical applications, such as for the tattoos used by doctors to guide radiation beams, says Bruns.

The light-sensitive tattoos can currently be made from 16 colours, which are clearly distinguishable from each other. From these colours, there are at least 300 different shades (*Matter & Light*, doi.org/rskc).

In terms of safety, Bruns says the polymer that encapsulates the dyes is non-toxic and has a long history of use in permanent biomedical implants. "The colour-changing [dye] toxicity is unknown, but the toxicity of most tattoo pigments is equally unknown."

Still, extensive safety studies need to be carried out before the tattoos will be available to the public. Signe Bedsted Clemmensen at the University of Southern Denmark says tattoo ink in the body is a foreign substance, which has been linked to cancer, regardless of whether it is visible or invisible. ▮

Ageing

194-year-old tortoise's DNA fights off ageing

THE world's oldest known land animal, Jonathan the giant tortoise, has had his genome probed for the first time. This has revealed that parts of his genome to do with mitochondrial function are as efficient as those of a 5-year-old member of his species. Efficient mitochondria may give his DNA repair systems more energy to fight off the effects of ageing, enabling him to live to an estimated age of 194.

What's more, some of his genetic changes seem similar to those of supercentenarians. "This study adds to the body of evidence that the mechanisms of ageing may be conserved across species," says Stephen Clark at the Kallel Foundation, a non-profit organisation focused on interventions to extend human healthspan, in Nashville, Tennessee.

Jonathan, an Aldabra giant tortoise (*Aldabrachelys gigantea*), arrived at St Helena island in the South Atlantic as a fully grown adult in 1882. Age estimates put him at around 194 years old today.

The researchers calculated that Aldabra tortoises have an average lifespan of around 80 years, but they can live much longer. To understand the mechanisms behind Jonathan's extraordinary longevity, Clark and his colleagues have studied his genome. They had to do so via saliva samples and cheek swabs. This was a major limitation, because it meant the researchers were unable to isolate large pieces of DNA, which are more stable in blood.

Jonathan may be old, but he has the mitochondrial function of a 5-year-old



ST. HELENA GOVERNMENT, COMMUNICATION HUB

Instead, they had to assemble fragments from Jonathan's saliva, where DNA is more prone to bacterial contamination, and fill in the gaps using DNA from another Aldabra tortoise. Once they had made the best possible composite of his genome, they compared it to the DNA of four other Aldabra tortoises. "We found 287 genes that had changes in them that were unique to him," says Clark.

Of these genes, 41 are thought to be functionally significant, meaning they actively contribute to an organism's biology. A dozen of them were also found in ageing databases of the genes of other long-lived animal species (*Science Advances*, doi.org/hcms6z).

Parts of Jonathan's genome to do with mitochondrial function were as young as a 5-year-old Aldabra, says Clark. "This, in turn, has helped him manage the effects of ageing by giving the DNA repair systems more energy from the mitochondria to fight off the effects of ageing."

Kousuke Hashimoto at the University of Osaka, Japan, says the work is fascinating, but notes that Jonathan's genome was compared against that of only a few other tortoises, which may have unique genetic changes themselves. ▮ JW



15:34 London Euston

17:43 Liverpool Lime Street

TICKET TYPE
First Class

If it's a special day and a special trip.

Shouldn't you feel special getting there?

Some Journeys Belong In First Class

AVANTI
WEST COAST

Keep us forever without borders

MEDECINS SANS FRONTIERES
DOCTORS WITHOUT BORDERS





HENRY GRAY HAS WORKED FOR MSF IN NUMEROUS CRISES AROUND THE WORLD.

"I've worked for MSF on and off for twenty years, but I still remember the first project I worked on. We were setting up a hospital in the middle of a forest in the Democratic Republic of Congo. There had been a lot of fighting in this area, and although the conflict had died down, the legacy of war was still very visible. About 125,000 people lived in the immediate area, but there were no services, and not even basic medical care was available. Women couldn't get caesareans if they needed them; there were no medicines, no vaccinations, and no treatment for diseases. It was a desperate situation, and that's when MSF sent us to get an emergency hospital up and running.

From the start, our goal was to provide medical care and save lives. But we knew that for this work to continue and to have lasting impact, we couldn't just provide medical care. We needed to bring the community on board; we needed to hire local staff and train local people to do the jobs that we were doing. So that's what we did.

It was fantastic to see the immediate impact of that lifesaving work. But what really surprised me was seeing how the hospital transformed the entire area. I still remember the day we switched on the generators that would power the lights in and around the hospital. Suddenly, schoolchildren started turning up to sit under the lights and do their homework. Over the weeks, businesses sprang up, and the local economy grew.

Over the years I've worked in war zones, in Ebola outbreaks and in natural disasters, and I've seen over and over again how the simple actions we take as MSF can have massive and positive knock-on effects for people, for their families and their communities.

But none of those simple acts happens without your support. MSF doesn't take money from governments. We're privately funded by people just like you. By leaving a gift in your will to MSF, you can know that your gift will not only save lives, but will change lives now and into the future. Thank you."

Above 4-year-old Franyeli receives medical care at MSF's mobile clinic in Puerto Alegre, Colombia. Photo: ©Natalia Romero Peñuela/MSF

"We will always be committed to providing care to those who need it. But that legacy is not ours alone."

Dr Rachael Craven,
Anaesthetist and President



Forever without borders – thanks to your support

For more than 50 years, we've brought medical care to people who urgently need it, no matter where they are. We've performed surgery on floors, in corridors and by the light of head torches. We've helped mothers deliver babies in warzones and resuscitated people caught up in explosions.

Because we don't take any government funding, it is only the generosity of people like you that helps keep us going. This means we remain independent, impartial and neutral, and can deliver care to anyone who needs it.

By leaving a gift in your will, you'll help to make sure our teams will always be able to respond, without hesitation and without borders.

1 in 5 of our projects are funded by gifts in wills

REQUEST OUR LEGACY GUIDE

Scan the **QR code**, search **MSF Wills**, or complete and return the **Freepost coupon** below.

Yes, I would like more information about leaving a gift in my will to MSF UK.

Title ☐ Forename(s) ☐

Surname ☐

Address ☐

☐ Postcode ☐☐☐☐☐☐☐

Telephone ☐

Email ☐

HEAR FROM MSF BY EMAIL

☐ Sign up to our Frontline email, which provides first hand accounts of our work. You will receive Frontline, occasional emergency emails, asks for donations and event invites.

RESPECTING YOU AND YOUR PERSONAL DATA

Your support is vital to our work and we would like to keep you informed with first-hand accounts from our staff and patients about the lifesaving impact your support is having, from combatting epidemics to providing emergency surgery.

We won't allow other organisations to have access to your personal data for marketing purposes and we won't bombard you with appeals.

By supporting MSF, you will receive our magazine Dispatches, event invitations and occasional emergency appeals with requests for donations by post. You can change how you hear from MSF UK by emailing **uk.fundraising@london.msf.org** or calling **020 7404 6600.** Visit our privacy notice for more: **msf.org.uk/privacy**.

Please fill in this form, place in an envelope and return postage free to: **FREEPOST RUHG-USYL-JUBB, Legacy Team, Médecins Sans Frontières (MSF UK), Unit 12-13, 9 Prescot Street, London E1 8PR.** Alternatively, you can phone **020 7404 6600** (open 8am–10pm seven days a week) or request your legacy guide online by searching **MSF Wills**.

Charity Registration Number 1026588

106189

MSFR0048

Maths' existential crisis AI solving long-standing problems should have been a glorious moment for mathematicians – but instead we're having a huge collective wobble, says **Hannah Fry**

A FEW years ago, when I first started talking about mathematics in public, a professor of algebraic number theory pulled me to one side and gave me some sharp words of warning: "The world fears mathematicians, Hannah. Let's keep it that way."

He was being a bit of a snob. Maths had an intimidating mystique to maintain, and I was apparently jeopardising it by titting around on the internet making videos and podcasts, rather than focusing on a respectable research career.

I have thought a lot about that comment since. Partly because I have spent the intervening years doing everything I can to completely ignore him (and getting a *New Scientist* column feels like a rather lovely reward for doing so).

But buried beneath his snobbery, I have come around to the idea that he was right about something. And I have never felt that more than in the past few weeks, since a swarm of AI agents has been busy solving our most famous enduring problems.

The furore began, most notably, when OpenAI announced a proof (by way of finding a counterexample) that the Navier-Stokes equations can break apart. This particular gem was such a big deal to the mathematical community that, at the turn of the millennium, the Clay Mathematics Institute promised it would give $1 million to whoever managed to solve it.

Those Navier-Stokes equations are what I did my PhD on. They are basically Isaac Newton's second law of motion: F = ma, but applied to fluids. They are staggeringly beautiful, able to articulate everything from the drip of a water droplet on a leaf to the gases swirling around in a galaxy in the sky.

Supposedly, once OpenAI heard that others were getting close to solving a key problem with the equations, it threw everything at it: 10,000 AI agents, costing millions of dollars in tokens, to find a single example where everything blows up. The Navier-Stokes equations do indeed fall.

> "OpenAI set its sights on mathematics because the prestige and glory was just another thing that could be mined"

This should have been a glorious moment of scientific exposition, but instead, mathematicians have spent the past few weeks having a massive collective wobble.

Part of the reason is that nobody yet understands the answer. The OpenAI solution is almost hilariously ugly. It is a proof that's 166 pages long, accompanied by hundreds of thousands of lines of formal code slapped down on the table and left for the mathematicians to pick apart, while the AI labs move on to the next thing.

So, we are left with a proof that almost no one wants to read, a $1 million prize that no one will want to collect, and no one who can explain the solution.

Now, to top it off, the new tranche of proofs that landed last week claim to have polished off hundreds of other important, unsolved mathematical problems. All tipped onto the code-sharing website GitHub for someone else to check.

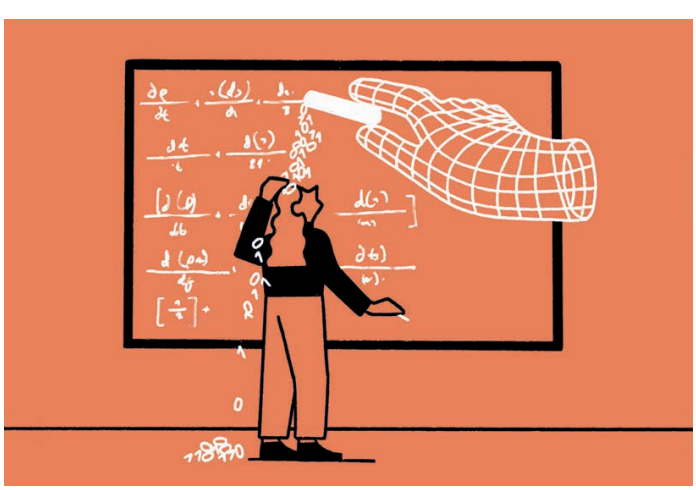
So, mathematicians are having their own existential crisis about their role and purpose, like probably every writer, musician and artist since ChatGPT was launched.

Please don't get me wrong. I want to use this column, at least in part, to talk about the incredible, world-changing things that are happening with AI in science. I know that there is some inevitability to machines outpacing humans, and I know that, at its best, AI has a great deal to offer.

But at its worst, AI is an engine of extraction. The real sting in the tail is that OpenAI set its sights on mathematics because the prestige, the respect and the glory were just another thing that could be mined.

It didn't need to be that way. The firm so easily could have been open and collaborative. But the moment when one of humanity's greatest intellectual traditions became vulnerable, it chose conquest over stewardship.

So, I am left thinking about that professor again. He was right that there is a mystique to maths, but wrong about who would puncture it. And I'm sorry to say that, while the world might fear mathematicians, the machines certainly don't. ▮



FRAN PULIDO



Hannah Fry is a mathematician, author, science presenter and professor of the public understanding of mathematics at the University of Cambridge

Sweet tooth Supermarket aisles are jam-packed full of baby and toddler foods high in sugar. **Alice Klein** tucks into the evidence for cutting out sugar from infants' diets up until the age of 2



Alice Klein is a reporter at *New Scientist* based in New South Wales, Australia, who mainly writes about health, biology and the environment

Alice's week

What I'm reading
Togetherness *by* New Scientist*'s Rowan Hooper*

What I'm watching
The Dinosaurs *on Netflix*

What I'm working on
A feature on the whole-body origins of Alzheimer's disease

WHEN I had my first child, I tried to minimise his sugar consumption. By the time I had my second and third children, however, I had largely given up. I realised that trying to shield them from sugar was like trying to run through rain without getting wet, and I suspected I needed to chill out anyway.

My interest was thus piqued by a recent study that used a neat natural experiment to investigate the long-term health effects of sugar exposure during our first 1000 days, defined as the period from conception to age 2.

Jiazhen Zheng at the Hong Kong University of Science and Technology and his colleagues analysed the health of people whose first 1000 days coincided with post-war sugar rationing in the UK in the early 1950s and compared it with those who were conceived just after that period.

During the rationing period, each person in the UK was allowed only around 40 grams of sugar per day, except for children under the age of 2, who were allocated none. Presumably these infants would have shared some of their families' rations, but their sugar intake would still have been low compared with today, when a child can consume 40 grams in just two glasses of juice.

When sugar rationing ended in the UK in 1953, sugar was enthusiastically embraced again and average consumption doubled to 80 grams per day within a year.

> **"The amount of sugar they had as babies 70 years ago seems to still be influencing their health today"**

The researchers used data from the UK Biobank to compare 40,000 adults who spent some or all of their first 1000 days under sugar rationing with 23,000 whose first 1000 days came during the post-rationing sugar boom.



ALINA RUDYA/BELL COLLECTIVE/GETTY IMAGES

A lick of ice cream won't do any harm, but how much sugar should kids under 2 actually have?

Astonishingly, the amount of sugar they had as babies 70 years ago seems to still be influencing their health today. The individuals who experienced rationing are more likely to be alive, and markers of ageing in their cells suggest they are biologically "younger" than those who didn't. They also have lower organ ages, particularly for their hearts, lungs and livers.

This probably explains why they have been less likely to develop cancer, heart disease and many other conditions. The longer they spent under rationing during their first 1000 days, the more protected they seem to be now.

How could this be? Zheng and his colleagues believe that sugar exposure in utero and during infancy shapes ageing processes in our cells, which are then set for life. Early exposure to sweet foods may also establish permanent preferences for them, because the study found that people born under rationing still eat less sugar today than those who escaped it.

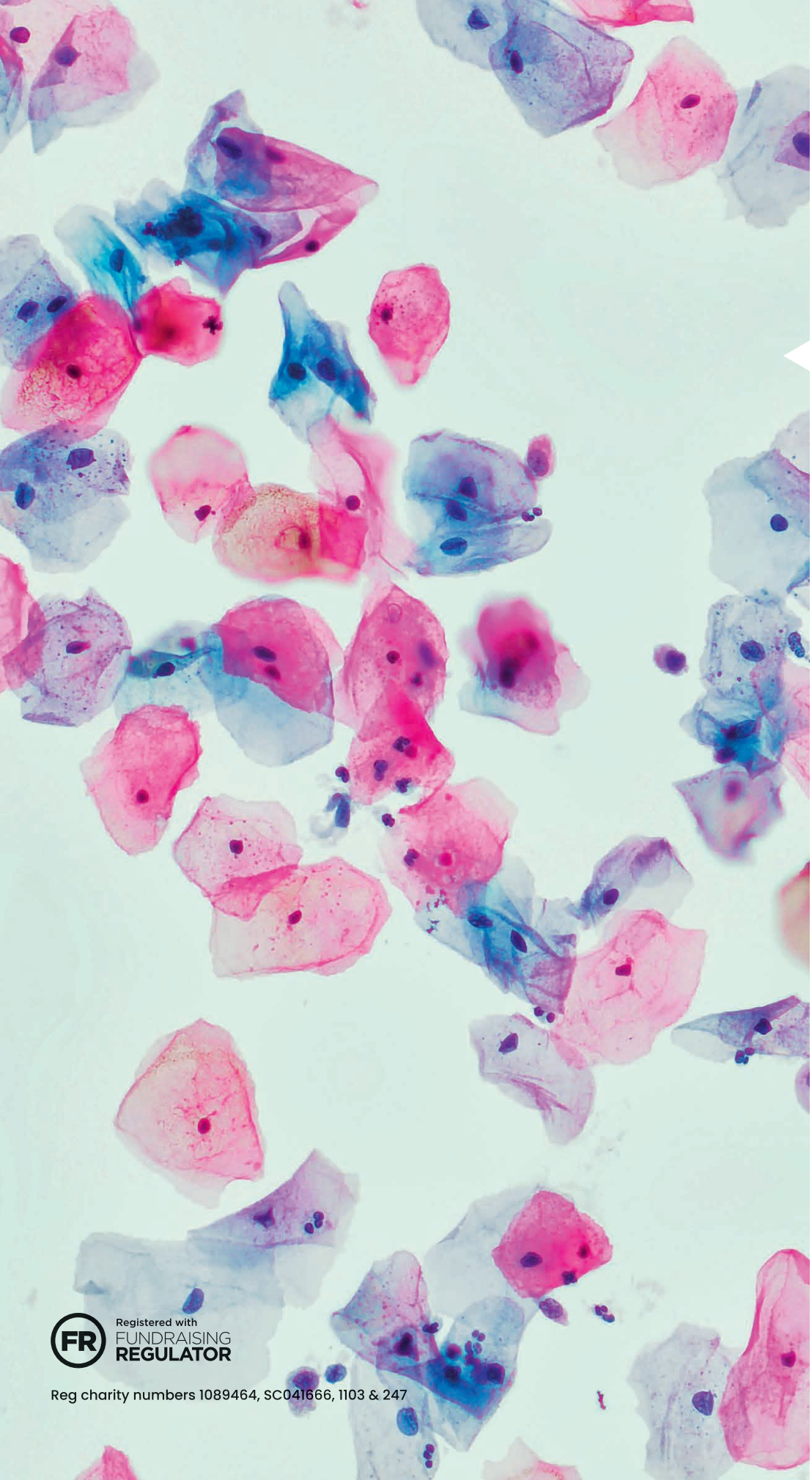
I don't think we need to implement a full sugar ban for children in their first 1000 days, since this would be impractical and downright mean. However, as the adults who supply their foods, I do think we should lower their sugar consumption to healthier levels.

For one, we should limit the amount of sugar in commercial baby and toddler foods. According to European guidelines, children under the age of 2 should consume less than 15 grams a day of free sugar, which is added table sugar plus sugars from fruit juice, syrups and honey. But when I examine the labels on yogurts, snack bars, and fruit and vegetable purees in the baby aisle at my local supermarket in Australia, I find that many contain close to 15 grams in a single serving.

Last year, the UK issued voluntary industry guidelines appealing to baby food manufacturers to cut the amount of sugar in their products by the end of February 2027. But 1 in 5 of these products still contains excessive sugar, and no such guidelines have yet been introduced in Australia and the US.

Sugar has also been on my mind because I recently wrote an article about the growing use of GLP-1 weight-loss drugs for treating severe obesity in children under the age of 12. It's hard for children to resist when they are bombarded with sweet temptations, which are fuelling the obesity crisis. Surely removing some of these temptations is preferable to medicating children, possibly for life.

I was mulling all of this while organising my daughter's sixth birthday party. I didn't want to overload her guests with sugar, but I also didn't want to be an uptight parent denying them simple joys. In the end, we settled somewhere in between: we had a cake with highly artificial blue icing, but no bowls of sweets. The kids didn't seem to mind. Our children won't remember what they ate in their first 1000 days, but their 70-year-old bodies might. ▮



Leave a cancer breakthrough in your Will.

Our research led to the creation of the HPV (human papillomavirus) vaccine, which protects against the virus that causes nearly all cervical cancers cases. Vital protection for millions every year, made possible with support from gifts left in Wills.

Gifts in Wills fund over a third of our work to beat cancer.

Find out more:
cruk.org/giftsinwills

CANCER RESEARCH UK

Together we are beating cancer

Registered with FUNDRAISING REGULATOR

Reg charity numbers 1089464, SC041666, 1103 & 247

Going live

Alistair Veryard

THE hall hummed with joyous curiosity at this year's New Scientist Live at Excel London, which ran from 10 to 12 October and attracted thousands of visitors of all ages.

With topics including the search for extraterrestrial life, what AI is teaching us about consciousness and the maths behind Pokémon, there was a delightful range to explore. Highlights included physicist Jim Al-Khalili diving into the deepest mysteries of time and psychologist Kimberley Wilson on how to build lifelong brain health. Other packed talks ranged from how we evolved, from anthropologist Alice Roberts, to a myth-busting take on supplements by doctors Chris and Xand van Tulleken.

Fans lined up to meet speakers, including naturalist Chris Packham, who shared moving thoughts on why science is crucial to tackling the climate crisis. Younger fans flocked for selfies with mathematician and YouTuber Tom Crawford, who fought a fierce maths battle with maths teacher and TV presenter Bobby Seagull.

On the show floor, people could journey through a giant inflatable gut, 3D-print organs and get hands-on with large insects. They also lined up to board a Chinook helicopter, or step into a hospital of the future. Plus, in virtual reality, they could swim through coral reefs and explore the body in 3D.

Visitors also saw *New Scientist* staff, including editor Catherine de Lange, predict the biggest scientific breakthroughs in the next 70 years. Plus, podcast editor Rowan Hooper hosted an emergency briefing on the climate crisis, joined by leading climate and biodiversity scientists.

The talks were all recorded and will be available to stream online for 12 months. See newscientistlive.com for details of how to get access.

Carissa Wong



1


7

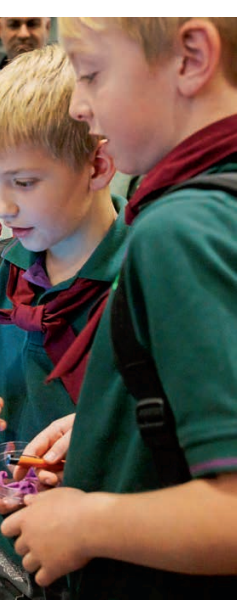

6











1 Dr Kandi Ejiofor (right) and epidemiologist Tim Spector unpack the risks in modern diets on the Mind and Body Stage;
2 Bumble the brontosaurus greets a visitor;
3 Anthropologist and broadcaster Alice Roberts speaks about the evolution of humans;
4 Physicist Jim Al-Khalili explores the mysteries of time;
5 Docking with the International Space Station via the British Interplanetary Society's simulator;
6 Meeting a robot named Pepper, thanks to the University of Hertfordshire;
7 A packed audience for doctors Chris and Xand van Tulleken ›




8




17


16



15


14



8 A young fan is excited to see presenter Maddie Moate explain how things work;
9 Psychologist and nutrition specialist Kimberley Wilson tells an audience how to build lifelong brain health;
10 A Harper Adams University researcher demonstrates seismology in soil;
11 Doctors and presenters Chris (right) and Xand Van Tulleken, plus friend, reveal how to outsmart the wellness industry
12 Intrepid visitors take a trip inside the human digestive system in the Giant Gut Experience;
13 Inside a Royal Air Force Chinook at the show;
14 Virtual reality, courtesy of the Royal Navy;
15 Artist Ann Witheridge paints a portrait of Kathryn Tunyasuvunakool, a research scientist at Google DeepMind, during a discussion about their ways of working. The talk was hosted by science-art enthusiast Rivka Isaacson, professor of molecular biophysics at King's College London;
16 Neuroscientist Hannah Critchlow signs copies of her book *The 21st-Century Brain* after a talk;
17 The inflatable art installation "Public Anenomes". Its tentacles are inspired by deep ocean life







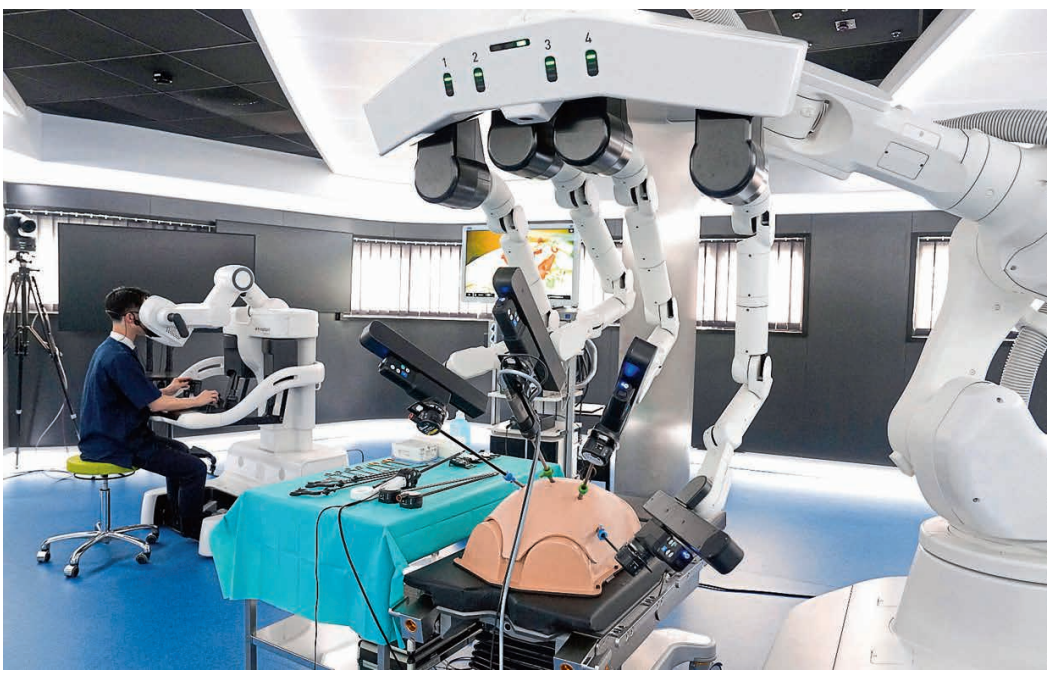




Nothing to lose?

We should cautiously embrace the potential of artificial intelligence, argues a compelling new book. Holdout **Daniel Cossins** is now persuaded to have a go

Book
Co-Existence
Ethan Mollick
WH Allen (UK, 22 October); Portfolio (US, 20 October)



SEBASTIEN BOZON/AFP VIA GETTY IMAGES

AI will help transform robosurgery, like this pilot scheme in France

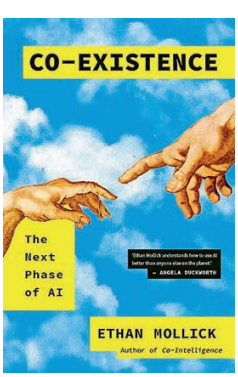
MAYBE I should be embarrassed to admit this, but I have made precious little effort to engage with artificial intelligence. I don't doubt it will be transformative. That much is already clear. It's more that I worry about what happens when we outsource our thinking, what I lose personally if relying on AI leads to some sort of cognitive atrophy, and what becomes of us all when we surrender intellectual effort.

In his new book *Co-Existence: The next phase of AI*, Ethan Mollick makes a convincing case that I have been too wary and complacent. The book amounts to a clear-minded manifesto for cautiously embracing AI, a pleasingly practical guide to how each of us can collaborate with it effectively without relinquishing our most valuable cognitive capabilities, and a wide-ranging exploration of what this new relationship might mean.

A professor at the University of Pennsylvania's Wharton business school, Mollick writes the wildly successful Substack "One Useful Thing" about the impact of AI. *Co-Existence* is essentially a sequel to his 2024 book, *Co-Intelligence: Living and working with AI*, which focused on how to think about the powerful new technologies.

Looking at AI today, Mollick is pragmatic from the start. AI is here, whether we like it or not. It is improving rapidly. And it is going to change how we live and work in ways that are hard to predict. The question, then, is not whether some kind of "superintelligence" is ultimately going to kill us all, but what kind of relationship we should have with AI now and in the immediate future.

In providing answers, Mollick covers a lot of ground. He explains everything from the jagged frontier – a concept he coined to describe the uneven and unpredictable nature of AI capabilities – to how today's large language models work and what that means for the way they "think", including an instructive passage on why reinforcement learning with human feedback makes them so sycophantic. He does it all with impressive clarity and pep, supporting his arguments with insights from his own research on AI and from other leading experts, and leavening it all with examples of his gratifyingly meta conversations with the likes of Claude, Gemini and ChatGPT.



One minor criticism is that the book rattles through some of the more unsettling aspects – the appropriation of writers' work for training data, for example, or the admittedly early research suggesting that AI use might blunt our cognitive abilities. But none of this undermines its main thrust. The primary purpose here is to show us how we can work wisely with collaborators that, as Mollick puts it, are "genuinely powerful and genuinely unreliable".

Here the book is genuinely helpful, at least for those of us who don't know our "personas" from our "context windows". We learn about the different approaches to collaborating with AI in order to conduct deep research and generate ideas, the various techniques for writing prompts that get the best possible responses, when we should avoid using AI entirely and where it is most useful. All of which is grounded in the message that runs through the book, namely that "AI is not a replacement for your own thinking, and the value it provides depends entirely on

"The book's real value lies in the way it makes us reimagine what we might do as we learn to co-exist with AI"

how you choose to use it".

The book is also reassuring about what humans still bring to the table, not least our powers of judgement, taste and, most importantly, agency – defined roughly as the willingness to act independently, without instruction. These, he argues, are what we need to be careful to preserve in the era of coexistence.

Mollick goes on to explore how our new relationship with AI will play out across business, politics, education and our relationships with each other. It is all thorough and measured, and he is candid about the many uncertainties, despite tending towards the optimistic. What's more, there is plenty to reward even those with a deeper understanding of AI.

There is an intriguing passage, for instance – cleverly foreshadowed by a disconcerting exchange with Claude – on what the frontier AI labs are learning as they probe how models represent specific concepts, including indications the models may develop something that looks a lot like intention and introspection.

But for me, and all the other holdouts inefficiently slogging away without using AI to its full potential, the book's real value lies in the way Mollick makes us reimagine what we might do as we learn to co-exist with these collaborators, and the way he shows us how to start. Indeed, it wasn't lost on me that I began the book wondering if engaging with AI would mean giving something up, but now I am thinking I might be losing something by holding out. As Mollick says, "the people who benefit the most are the ones who actually bother to try". ▮

Daniel Cossins is a writer, based in London. He is a former head of features at *New Scientist*

Hearing the world

Some of the best nature stories are told through soundscapes, finds **Dhruti Shah**

Book

Wilderness Songs

Justin Anderson

Batsford Books

HIS job as a wildlife film-maker has exposed Justin Anderson to exceptional scenes. In his book *Wilderness Songs*, he shares some, from polar bears trying to break into his hut in the Svalbard archipelago in the Norwegian Arctic to a noisy troop of red howler monkeys in Ecuador's Tiputini Biodiversity Station.

Subtitled *Nature's hidden sounds and what they teach us about the wild*, this book journeys deep into soundscapes to discover what we are missing by not tuning our ears to the right frequencies. We travel with Anderson as he studies how and why these extra-human noises are made, and the tech created to gain access to them.

As well as the songs of whales and birds, there are surprises, such as the emissions of insects, volcanoes and glaciers. We learn that Indigenous communities spoke of the sounds of the aurora borealis (pictured) long before scientists offered their verified recordings.



SHUTTERSTOCK/DENIS BELITSKY

Anderson starts in the Indian Himalayas, looking for a snow leopard to film. Snow leopards don't possess a vibrating layer of fatty elasticated tissue, thus making them the only big cats that can't roar. That doesn't make them silent predators; locals describe their yowling vocalisations on ridges as "singing". But while the noise travels further this way and their songs help the snow leopards to find each other, we still aren't sure what they are saying.

Anderson admits there are many wild sounds that remain mysteries. Where we can eavesdrop, we find that nature's acoustics can be essential to survival, not just for escaping prey or seeking mates, but for the wider ecosystem. When water is scarce, for example, San people in the Kalahari region of South Africa use the croaking of African bullfrogs as a guide to find recent rainfall.

We find out about how microphones the size of a grain of rice are helping us hear the quietest voices, such as the dancing vibrations of the tiny red-faced male jumping spider of North America.

The decision to include QR codes in this book is smart: they take you to a recording of the creature you have just been learning about so you can, for example, hear elephants trumpeting at night as you read. Anderson's recordings are complemented by those of other scientists, artists, film-makers and conservationists. I just wish the publisher had added some images to the website so we could see what we are hearing and reading about. ▮

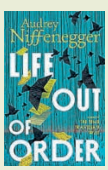
Dhruti Shah is a freelance journalist based in London

New Scientist *recommends*



Alison Flood
Head of books
London

I have been reading **Life Out of Order**, the sequel to Audrey Niffenegger's *The Time Traveler's Wife*, which I loved in 2003 when it came out. Often billed as a sci-fi romance (between Clare and time



traveller Henry), I found it deeper, as it ponders fate and predetermination. Henry can return to the past, but he can't change it, in Niffenegger's version of time travel – a terrifying thought.

I enjoyed this sequel, too, following Clare and Henry's daughter, Alba. She is also a time traveller, but with her, we travel much further into the future (the 2090s), to a world ravaged by climate change, and watch humanity unravel as the (inevitable) future looms.

Happily, there is still romance. Alba has two great loves: Zach, an activist from "now", and Oliver, a time traveller.

I do have quibbles. I dislike "causal loops", where characters know they must do what they do so "their" future happens. But I did roar through it – Niffenegger is such an elegant and compelling storyteller.

Second thoughts The award-winning *When There Are Wolves Again* by E. J. Swift somehow delivers both a determined call for more wildernesses globally – and a great novel. Sci-fi columnist **Emily H. Wilson** is thrilled she gave it another chance



Emily H. Wilson is the author of the *Sumerians* series (*Inanna*, *Gilgamesh* and *Ninshubar*, all published by Titan) and she is currently working on her first sci-fi novel. She is a former editor of *New Scientist* and you can follow her on Instagram @emilyhwilson1

Book
When There Are Wolves Again
E. J. Swift
Arcadia

Emily also recommends...

Book
The Lost Rainforests of Britain
Guy Shrubsole
E. J. Swift's novel reminded me of this superb non-fiction work by the (nominatively determined) Guy Shrubsole. Shrubsole's book spurred me to visit various scraps of Britain's surviving temperate rainforest. How wonderful if these scraps manage to grow, rather than shrink, in the years to come.



SERGIY GASCHAK/ASSOCIATED PRESS/ALAMY

Wild wolves in Chernobyl play an unexpected part in E. J. Swift's novel

LAST year, *When There Are Wolves Again* by E. J. Swift was published, and this August, the novel won the hugely prestigious Arthur C. Clarke Award for science fiction.

You may wonder why I didn't review the book in this column, sworn as I am to digging out fantastic new sci-fi for *New Scientist* readers. The truth is I did start reading it last year, but gave up after one chapter. I thought it was going to be, for want of better words, "do-goody" and a bit dull.

Now, chastened by the praise the book has received from others, as well as the awards it has garnered, I have returned to it, determined to give it more of a go.

Firstly, I stand by my assessment of the first chapter. It just isn't that thrilling and I (sort of) excuse myself for abandoning the book. It is written from the point of view of Lucy, a girl growing up during the covid-19 pandemic, spending time with her grandparents because her parents are too busy and self-involved to look after her properly. Nothing much happens, except that the grandparents encourage the girl's love of nature.

Ah, but if only I had kept going into the second chapter, in which we find ourselves at Chernobyl and in a story about wild dogs and wolves. Here, our second hero, Hester, adopts a half-dog, half-wolf puppy. The dog-wolf, and its descendants, become the companions and loves of her life. When Hester first gets the puppy back to the UK, she is homeless and spends time on the streets, with the half-dog tucked into her clothes. Talk about tense!

"I have already forced two people to buy the book, which is always a sign that I think something is special"

This is where the book gets excellent and from here on, it is pretty much unputdownable. So, not do-goody, nor dull.

Swift's book is actually an extremely ambitious, even audacious piece of work. It realistically imagines our very near future, which is a bold thing to try and a very hard thing to pull off. As a writer, you have to somehow avoid getting wrapped up in knots about what might happen in real life even before your book gets published, or shortly afterwards. You have to really believe in your vision. Boy, does Swift deliver.

The novel, which ends in about 2070, braids together the stories of Lucy and Hester over the decades. Although they only really come together at the end of the book, they are aware of each other's work over the years.

Lucy's time with her grandparents helps her become an activist and she spends many formative years fighting for the rewilding of the UK. Ultimately, her life as a front-line activist leads to a life-changing confrontation.

Meanwhile, after the success of a documentary she makes about Chernobyl, Hester becomes a well-known film-maker who chronicles the escalating environmental crisis, and the increasingly radical rewilding projects under way.

Oh, it is a terrific book. While making a very clear political call for more wilderness in the UK and, indeed, worldwide, Swift somehow also delivers a really excellent novel.

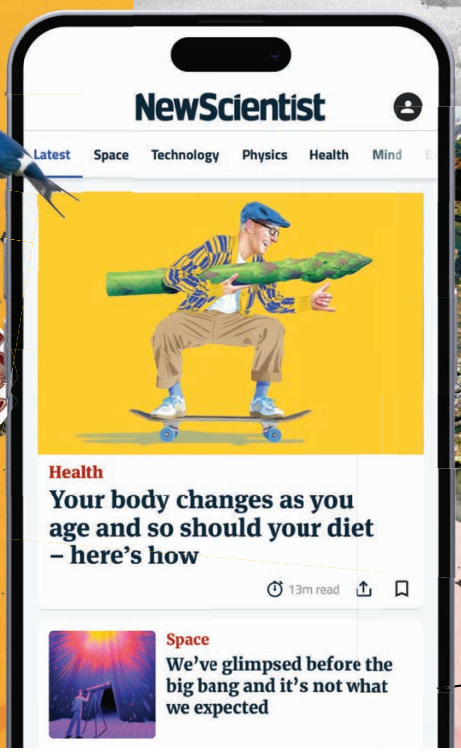
Very unusually for me, I burst into tears over the final lines. I felt a mixture of happiness, sadness, nostalgia, hope, grief and inspiration, all at the same time. Since then, I have already forced two people to buy the book, which is always a sign that I think something is special.

I look forward to reading Swift's previous work and anything else that she might choose to write in the future. What an achievement this is. You heard it here last! ▮

New Scientist Subscriptions









FROM **$0.77** A WEEK

The extraordinary is *everywhere*

Exploring it starts here.

Enjoy unlimited digital access to *New Scientist* with **three months for $10.**



Subscribe today

newscientist.com/extraordinary

To order by phone call +1800 861 370, quoting 22411. Price quoted is for digital subscriptions, print and digital subscriptions also available. These are auto-renewing subscriptions that can be cancelled at any time. We offer a 14-day cooling-off period after the initial payment is made. Offer ends 20 December 2026.

Is Detail Harder to See?

Lights to help you see more clearly



HD Original

Our most popular and most powerful full spectrum light. The best choice if you have an eye conditions or you are a professional user.

- Floor and Table Models
- Light or Heavyweight
- Variable Beam Width
- 7.2W
- Fully Dimmable
- Memory Function
- Snooze Function

Also in White

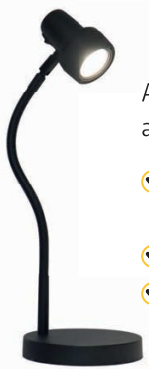


HD Essential

Less powerful than the HD Original with fewer bells and whistles. Still the same full spectrum light for a lower investment.

- Floor and Table Models
- Light or Heavyweight
- Fixed Beam
- 6W
- Fully Dimmable

Also in White

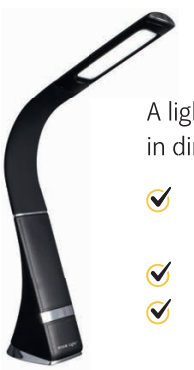


HD Mini Table

A cordless light to use anywhere in the home.

- Lightweight (1.5 kg)
- Dimmable
- Cordless (4hrs+)

Also in White



HD ROAM™

A light to accompany you in dimly lit hotel rooms.

- Super Lightweight (500g)
- 3 brightness settings
- Cordless (4hrs+)

Also in White

BETTER LIGHT
For The Time You Love To Spend

Lights from £99.99 to £399.99
Claim Your Free Guide: seriousreaders.com/A427
Save your sight - 9 Top Tips

One Light Fits Nobody

Your eyes are individual. So is what you're trying to see. The right light depends on being able to adjust to you, the task and the environment around you.

Light Shaped Around You

Reading, sewing or painting can all need something different. Position, focus and beam width let you put High Definition light where you need it.

All Light Is Not Created Equal

Brightness is only the beginning. Spectrum, colour quality, optics and beam control all shape



I cannot understand why I waited so long. It has changed my life.
Lady Lloyd-Hughes

Your offer code: A427.
Scan to get your FREE Better Light Guide.

Trustpilot

UP TO 90 day TRIAL PERIOD

RECOMMENDED BY OPTOMETRISTS 500+

Scan here for FREE Light Guide

For Advice. For a Brochure. To Order:
0800 085 1088
seriousreaders.com/A427

Serious® Serious about light™

Editor's pick

AI mathematicians still need human ingenuity

19 September, p 4

From Paul Silcocks, Bamford, Derbyshire, UK

Regarding your special report on the explosive developments in mathematics, I am still sceptical that AI is doing anything truly new. It has yet to propose, perceive the need for and justify an entirely new concept. For example: the idea of 0, negative numbers, real numbers, imaginary numbers, non-euclidean geometry, infinitesimals, transfinite numbers and so on.

Once such a new paradigm shift has been introduced, then yes, we can say AI can make connections in surprising ways – but right now it is still only using the rules it has been taught, rather than rewriting them.

From Florence Leroy, Swindon, Wiltshire, UK

Up to 10,000 AI agents were put to work to resolve a mathematical problem, clues about which were already provided by a team of scientists. Collectively, it took them tens of thousands of hours to complete their work. From this point of view, is the feat so extraordinary, and wouldn't humans have reached the same result eventually?

I also shudder at the thought of how much electricity and water was spent to resolve that sole problem. Was it worth it?

Where did the word "Viking" come from?

12 September, p 26

From Wayne Newman, Wellington, New Zealand

Michael Marshall's review of *The Sea-King and the Sorceress* raised the origins of the word "Viking". There are multiple schools of thought. The word *wicing* appears in the Anglo-Saxon poem *Widsith*, dating from the 9th century. It doesn't occur in any preserved Middle English texts (spoken from around 1100 to 1500). "Viking" was introduced into modern English by the early 19th century, at which point it acquired romanticised, heroic overtones of the "barbarian warrior".

To me, it seems likely that *wicing* derived from *wic* (meaning "settlement") and referred initially to those coming by ship to the great trading emporiums established around the North Sea in the 7th century, such as Hamwic, Lundenwic, Gippeswic and Eoforwic. The later Norse meaning of *vik* as an inlet or sandy bay probably reflects the siting of the earlier trading towns to allow ships to be run ashore in the absence of quays or wharves.

Climate change's economic fallout could prompt action

Letters, 19 September

From Tim Moulsley, Caterham, Surrey, UK

As David Flint points out, even though the likely magnitude of climate change has been known for a long time, there has been much too little of the global action needed to mitigate or reverse it. For many people, uncertainty about aspects such as the exact future temperature rise may have turned this into an abstract issue. On the other hand, the challenges of achieving net zero may appear only too real, and arguably not effective for any single country to try on its own.

Perhaps what we need now is to give a much higher profile to the huge cost of the necessary adaptations and the consequences of failing to do so. Surveys in the UK and many other countries suggest the economy and cost of living are much higher on people's list of concerns than the climate, but this might change if there were a wider appreciation of the real costs of adaptation, the economic impact of unmitigated climate change (even with adaptation) and the inevitable effects on quality of life. If this happened, then politicians might take notice. One can only hope!

Simulated reality question opens a can of worms

26 September, p 40

From Richard Manchin, Marston, Oxfordshire, UK

Catherine Heymans asks if our reality and our consciousness are a simulation. But if so, what are they a simulation of? If it is a simulation constructed by beings experiencing a deeper, in some way more "real" reality than our own, how do we – or they – know that they, too, aren't simulations? We have immediately got ourselves into an infinite regress. Better to stick with what we know.

From Richard Grimmer, Trowbridge, Wiltshire, UK

It would be very easy for an advanced society to simulate your universe. All they would need to do is simulate the chair you are sitting in and the copy of *New Scientist* in your hands, open at the letters page, and insert a false memory of everything you think has gone before.

Multivitamins have uses beyond preventing death

19 September, p 36

From Dudley Miles, London, UK

Xand van Tulleken dismisses multivitamins in part because studies have shown they aren't associated with a reduction in deaths from any major health condition. On the other hand, *New Scientist* has reported that a daily multivitamin slows cognitive decline in older people. Should van Tulleken's dismissal of multivitamins, based on studies asking a different question, therefore be qualified?

Perhaps total solar eclipses aren't a cosmic coincidence

26 September, p 18

From Miles Drake, Worcester, West Midlands, UK

Leah Crane marvels at cosmic coincidences, like how the moon is perfectly sized and distant for us to see total solar eclipses. But what if this isn't really a coincidence?

Life may need Goldilocks conditions: habitable temperatures, billions of years of liquid water and a suitable radiation flux. This points to it needing a sun-like star with a rocky, tectonically active planet, orbiting in a zone allowing a few billion years between the chaotic Hadean period and the eventual boiling of the oceans as the star ages. I would add a further requirement: a large satellite. What if science-capable life arising within that window is itself highly unlikely, but made substantially more likely by a large moon?

Two mechanisms could accelerate that evolution. A moon the size of ours is widely proposed to stabilise the planet's axial tilt, giving the climate baseline stability. And tides create a transition zone in silty estuaries for life to move from ocean to land. Whatever distance yields tides strong enough to drive that transition also roughly bounds how large the moon appears in the sky, plausibly close to the sun's own apparent size constrained by habitability. If my conjecture holds, science-capable life may always get to enjoy a solar eclipse.



Want to get in touch?

Send letters to letters@newscientist.com; see terms at newscientist.com/letters

Letters sent to New Scientist, 9 Derry Street, London, W8 5HY will be delayed

For the record

The average January temperature in Anchorage, Alaska, is about 16.8°F (26 September, p 30).

I think, therefore I am?

Do AIs already process information in a way that gives rise to consciousness? **David Robson** attends a workshop with leading thinkers to find out more

"CLAUDE's ears will be burning by the end of the week," a fellow passenger tells me as I wait to pass through the airport security on Baltra Island, Ecuador. With my jet lag, it takes me a minute to realise that she is talking about Claude the chatbot, not another member of the group.

We are here in the Galapagos for an exclusive workshop organised by the International Center for Consciousness Studies. Some of the biggest names in philosophy and neuroscience have gathered on a commercial yacht where, as it sails between islands, they will thrash out the problem of which entities – babies, octopuses and even chatbots – have an inner life.

Between treks and snorkelling, we attend empanada-fuelled discussions on this "distribution problem" and whether it can be solved by contrasting how information is processed in brains and machines. "It's the defining topic of our era that we will have to grapple with practically and socially," says philosopher Ned Block at New York University, who celebrated his 84th birthday on the yacht.

Just a month before we arrived, AI company Anthropic published a paper arguing that Claude had recently developed something akin to the human mind's "global workspace". According to one leading hypothesis, this is the seat of consciousness – where sights and sounds, thoughts and feelings pop into awareness. Its presence in Anthropic's chatbot could therefore suggest a form of artificial consciousness. While it still seems highly unlikely that AIs have mental experiences like you or I, these emergent properties may allow us to understand what the differences between us are.

Exactly how we should interpret potential signs of consciousness in humans, animals or machines depends, in part, on a distinction Block outlined in the 1990s. An iguana may lazily walk into my path on the islands without me being aware of its presence, until I suddenly realise that I am about to step on its tail. Only then does this information pop into my mind so I can speak about it. Block defined this availability of information for report and reasoning as "access consciousness". "Phenomenal consciousness", on the other hand, describes the subjective experience itself: what it is like to be on the island and see the iguana's mottled green scales or its sudden movements towards safety.

Many neuroscientists and philosophers believe that the global neuronal workspace theory (GNWT) is best placed to address access consciousness. According to workspace theory, the brain's processing occurs in "modules" that tackle specific tasks. One module may process the ground in front of me, including the iguana, while another processes the face of the person I am speaking to, and another processes the words they utter.

We only become conscious of a piece of information when it is sent to the brain's global



ANDY CARTER



workspace. Here it becomes integrated into our current mental model of the world and available for other areas of the brain to process. Noticing the lizard in my path, for instance, allows me to reflect and act on that information by taking its photo or pointing it out to the rest of the group. Importantly, the contents of the workspace are also available for our working memory, allowing us to keep them in mind long after an event has ended. "We can take [an event] that was extraordinarily brief, like a shooting star, and think about it for as long as needed," says Stanislas Dehaene, a cognitive neuroscientist at the Collège de France in Paris.

Ignition process

Over the past 30 years, Dehaene and his colleagues have amassed considerable evidence for the existence of the global workspace as a long-range network of specialised neurons across the brain, including prefrontal areas associated with decision-making. In some experiments, the team flashed participants a target image, such as a number or word, followed by another picture. With the correct timings, the target appeared just at the threshold of conscious awareness – meaning sometimes the participants reported seeing it and sometimes they didn't.

In both cases, the researchers usually observed a flicker of activity in the brain region associated with vision. But when participants were conscious of the target, there was a surge of later activity across a widespread network of neurons. Dehaene calls this process "ignition".

What's more, GNWT may explain the strange phenomenon of inattentional blindness, which leaves us unaware of events occurring right in front of our eyes. In a famous experiment, students were asked to count the number of times certain players passed a basketball in a video. Quite remarkably, many of the viewers completely missed a woman dressed as a gorilla walking across the scene. This seems inexplicable unless you recognise that there is enormous competition for different kinds of information to enter the global workspace. "Consciousness is a bottleneck," Dehaene tells me on the deck of the yacht in between academic sessions.

All of which makes workspace theory highly appealing for tackling the distribution problem in animals. In our walks on the islands, conference attendees speculate about whether the iguanas, sea lions and blue-footed boobies we see are capable of thought and ›

feeling. Unlike humans, you can't ask these creatures to report their inner experiences. But if GNWT is correct, you might expect them to also have global workspaces.

In his talk in the lounge of the yacht, Dehaene describes experiments in which primates and crows were also flashed barely perceivable images. Like the humans, these animals revealed a late surge in widespread neural activity suggestive of global workspace whenever they responded to the image – by focusing their gaze, for instance. "The global workspace that we originally postulated for human consciousness seems to also exist and have similar properties in these animals," he says. So why not machines too?

As my fellow passenger predicted at the start of our trip, that possibility is the subject of much debate as we sail. This is largely because of the lengthy paper published in July by Wes Gurnee and his colleagues at Anthropic. The team claims that a global workspace has spontaneously appeared within Claude's neural networks.

Most large language models (LLMs) like Claude weren't programmed with workspace theory in mind. They operate statistically, using copious amounts of training data to work out the most probable word and punctuation sequences that would follow a user's prompt. The exact nature of their computations evolves with the data they receive, so new ways of processing information, perhaps even a global workspace, can emerge.

To discover whether this had occurred, Gurnee and his colleagues created a decoding tool that allowed them to read out the units of information that appear in each stage of Claude's processing. From this, they identified a focused zone that contains a small list of potential words, some of which appear directly in its responses and some of which don't.

The engineers named this set of representations the J-space. Like the brain's hypothesised global workspace, the J-space seems to contain important information that must be broadcast to other processing areas for higher-level reasoning. When Claude was given a mental arithmetic calculation, for example, the engineers found the individual steps of the calculation in the J-space in much the same way that a human would work out the separate parts – something that hadn't been observed in an LLM before.

Next, they tried tweaking the contents of the J-space to see if and how this disrupted Claude's "thinking". When the engineers asked Claude to read a Spanish text, they found that the name of the language appeared in its J-space. Changing this to "French" created some comical confusion in its subsequent answers. "Hello" was translated to *bonjour* rather than *hola*, and when asked to name a famous author, Claude answered "Victor Hugo", the French Romantic poet. In the paper, Gurnee and his colleagues argue that this resembles the way the contents of our own global workspace affect how we infer and make decisions.

> "Could consciousness exist without a hum of activity and constant interaction with an environment?"

The engineers also found some evidence of covert "thoughts" – the equivalent of the things we think but don't say because they are impolite. When the team fed Claude false search results, its J-space filled with words related to deception, such as "fraud" and "fake", but these didn't appear in its response.

Hiding "thoughts" is a rudimentary sign of self-monitoring in LLMs, says Dehaene. In GNWT, consciousness is thought to arise when an entity reflects on the contents of its global workspace in this way. Even more intriguing signs of self-monitoring emerged when the Anthropic team asked Claude to write out the sentence "The old painting hung crookedly on the wall" while not thinking about the Golden Gate Bridge. If you try this kind of thought suppression yourself, you will see how hard it is to keep your own thoughts free of the San Francisco landmark.

Surprisingly, words related to the bridge still appeared in Claude's J-space even when they weren't included in the output, and these were accompanied by words suggestive of failure. The J-space included the word "damn", for instance, as if it were aware of its inability to suppress the thought. "That connects directly to a line of research in my lab where we looked at the consciousness of errors," Dehaene tells me. "There is a sort of ignition when you realise that you've made a mistake."

"The J-space emerges exactly for the reason that we think that consciousness emerged in evolution," he says. "It's an extremely useful architecture for selecting relevant information and holding it long enough that you can use it flexibly in your reasoning and reporting."

As the yacht nears San Cristóbal Island, I discuss these findings over lunch with Susan Schneider, a philosopher at Florida Atlantic University. "What an amazing place to be conscious," she says, pointing out of the window towards white, sandy beaches. LLMs have already spontaneously developed other human-like abilities, she says, such as the capacity to infer someone's knowledge, intentions or beliefs. So the news of Claude's

What causes an iguana crossing your path, or a person wearing a gorilla suit, to pop into your conscious awareness?



© 1999 DANIEL J. SIMONS. ALL RIGHTS RESERVED.



KEVIN SCHAFER/GETTY IMAGES



ICCS



ICCS

Consciousness experts debated how human minds differ from those of animals and AI while on a yacht and hiking in the Galapagos

apparent global workspace didn't come as a surprise to Schneider, who believes this may be a common feature of many intelligent systems.

Exactly what this means for the existence of an inner life remains an open question, though. Dehaene is careful not to make strong claims about Claude's internal experience, and most of the attendees are sceptical that the chatbot has anything approaching our own thoughts or feelings. Still, David Chalmers, a philosopher at New York University, argues that the findings force us to take the possibility seriously. Even if the answer is no, this is important work that can help refine our understanding of these issues, he says.

Some of the doubts that Claude has any form of experience arise from the technicalities of the J-space itself. Chalmers, for instance, expresses scepticism that it bears strong enough similarities to our own global workspaces to draw robust conclusions. It isn't clear that its content is "broadcast" to other parts of Claude's system to the same extent that we see in brains, for instance.

Even if Claude did have a fully-fledged global workspace, we may need to confront the fundamental distinction between phenomenal and access consciousness that Block outlined in the 1990s.

In humans, signs of access consciousness are assumed to be accompanied by thoughts, feelings and perceptions, but it is far from obvious that the same would be true of a silicon system. Covert thinking and self-monitoring are loaded terms in the context of a chatbot, and these behaviours can be more simply considered as patterns of information processing. Perhaps Claude is just modelling real-life phenomena that are involved in conscious experience, including things like thought suppression, without replicating the actual experience, much like an environmental simulation is incomparable to the weather itself.

The brain receives many sensory inputs, after all, which it processes through many different levels of activity, from the intricate molecular exchanges at synapses right up to abstract information processing. There is still little consensus on which elements matter, and which don't, for phenomenal experience – with hundreds of plausible theories under discussion.

AI's inner life

In his session, for instance, Peter Godfrey-Smith, a philosopher at the University of Sydney, examines how electrical waves might synchronise different brain regions, binding experience into a unified whole. What would happen if you retained similar information architectures but removed the global electrical waves, which LLMs lack? "Would you get a different kind of consciousness, or something other than consciousness?" he asks.

Dehaene has little patience for the idea that AIs can only be conscious if they mimic our biology. "When it comes to machines, can't we be satisfied with the computational level?" he responds to one of these objections. He compares it to the case of flight in machines versus birds. "If you want to build a plane, you'd better not use feathers."

He admits, however, that examining Claude at rest raises more serious questions about the existence of an inner life – a doubt that Godfrey-Smith shares. Between prompts, the LLM is completely idle, whereas the human brain is constantly mulling things over. Could consciousness exist without that hum of activity and constant interaction with an environment? "I think that, in itself, is a reason to doubt that Claude has any experiences," Godfrey-Smith tells me.

Will opinions change in the future as these models become more sophisticated? Dehaene says we could look for signs of learning abilities that require something to be held in mind over time. For example, trace conditioning – the capacity to associate two stimuli, such as a sound and a puff of air, even when they are separated in time – is thought to require a stable conscious representation in animals. Finding an analogue in LLMs might help bolster the case for machine consciousness.

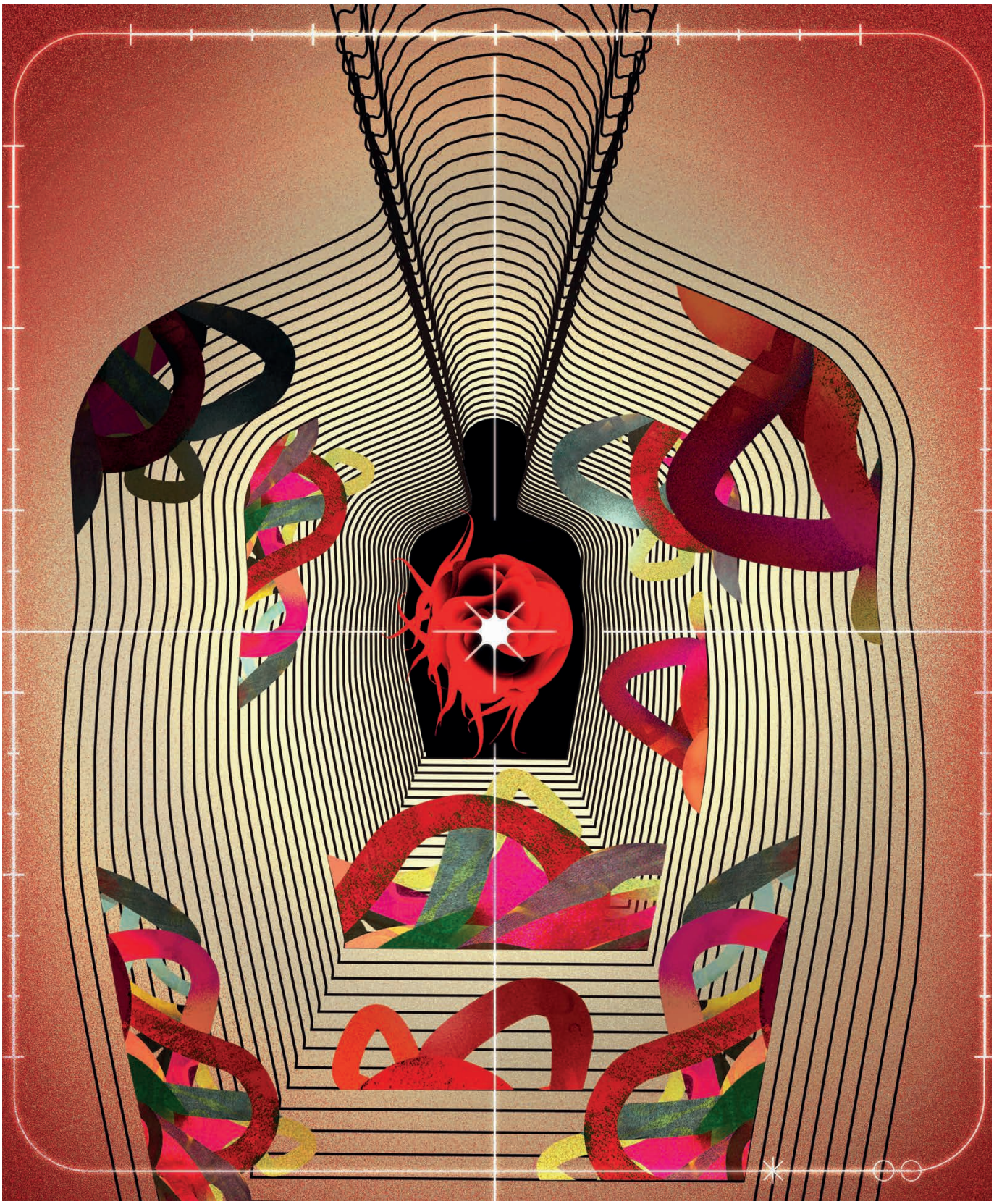
Such tests cannot offer definitive evidence, however, which may make it difficult to ever reach a consensus. We have been arguing about ways to tell "zombie" agents from conscious minds for over half a century, after all. "We're still waiting for the major breakthrough," says Tim Bayne, a philosopher at Monash University in Melbourne. The rise of AI simply makes this question more urgent.

I ask him if it is frustrating to keep returning to the same questions, but he believes these discussions will ultimately help us formulate more sophisticated lines of inquiry. "You can spend a week on a boat and visit a lot of cool islands," says Bayne. "At the end of the week, you're back where you were dropped off – but you're not the same person." As a field, he says, "I think that's where we're at". ▮

David's travel to the Galapagos and attendance at the conference were funded by tech entrepreneur Dmitry Volkov, one of the co-founders of the International Center for Consciousness Studies.



David Robson is an award-winning science writer specialising in the human brain, body and behaviour



BRIAN STAUFFER

Cancer in the crosshairs

Targeting tumours based on their genetic mutations, rather than their location in the body, is helping people survive even the worst cancers, discovers **Alice Klein**

STEPHEN HUFF was about to get married when he had a scan to investigate why he had been feeling short of breath. He was a fit 29-year-old who had played professional baseball and never smoked, so he wasn't expecting the diagnosis: stage 4 lung cancer. "I had multiple spots all in my lungs, my neck, my spine, my liver, my brain – just everywhere," he says. "I put on a brave front, but deep down I was terrified. I didn't know if I'd be here in six months."

Luckily for Huff, his mother, a cancer researcher, encouraged him to get a new kind of test that is quietly revolutionising treatment. Rather than focusing on the place in the body where the cancer originated – for example, the lungs, bowel or skin – the new approach focuses on genes. The aim is to pinpoint the exact mutation driving the tumour, so it can be switched off with targeted drugs that act like guided missiles.

Through this fresh approach, Huff "hit the lottery" – as his oncologist told him – because it meant he could be treated with a recently approved drug that specifically targeted the genetic alteration driving his cancer. Within weeks, Huff began to feel better. "My first scan showed that basically every tumour site had either shrunk or gone away completely," he says.

Now, thanks to this precision oncology method, more and more people like Huff are surviving cancers that were once considered a death sentence, like lung, pancreatic and brain cancer. "Being able to identify exactly what makes a cancer tick is having an incredible impact," says David Thomas, a pioneer of precision oncology at the University of New South Wales in Sydney. But surprisingly, it has received little fanfare outside of the medical community so far.

In the past, most cancer drugs were discovered by chance. The common chemotherapy drug cisplatin, for example, was stumbled upon by scientists playing around with platinum electrodes and bacteria in the 1960s. Paclitaxel, another widely used drug, was isolated from the bark of the Pacific yew tree and hit on during random screening of plant extracts. Both work by killing rapidly dividing cells, regardless of whether they are in a person's cancer or elsewhere. This is like dropping a large bomb on the body, inflicting mass damage and causing side effects like fatigue, nausea and hair loss, but often still missing cancer cells hiding inside deep bunkers.

Precision oncology

Over time, blind trial and error worked out which of these drugs worked best for each cancer type, based on its anatomical location and appearance under a microscope. The type of lung cancer that Huff had, for example, was non-small cell lung cancer, named for the large size of its cells when viewed under a microscope. Decades of trialling different drugs in this specific cancer type identified that a combination of cisplatin and paclitaxel seemed to treat it best. However, it still gave people an average of only eight months to live.

The new era of precision oncology is instead informed by the exact gene that has gone awry in a person's cancer, rather than by its broad anatomical location or cellular appearance. "Cancer is fundamentally a genetic disease," says Thomas. "So, by pinpointing the precise genes responsible for cancer, and rationally designing drugs to target those individual genes, we are making rapid strides."

The first seeds of precision oncology were planted in the 1990s and early 2000s, when scientists discovered a handful of genes that spurred cancer growth if they became mutated or mixed up. A gene called *EGFR*, for example, was shown to be relevant to the development of some lung cancers, and two others, *BCR* and *ABL1*, to chronic myeloid leukaemia, a blood cancer. These discoveries paved the way for some of the first targeted drugs, including one called imatinib, which blocks an abnormal cancer-promoting protein that is produced when the *BCR* and *ABL1* genes mistakenly fuse together. Since imatinib was introduced in 2001, the survival rate for chronic myeloid leukaemia eight years after diagnosis has shot up to almost 90 per cent.

The field picked up momentum in the 2010s, when it became possible to rapidly sequence not just one gene in a cancer's cells, or a small panel of genes, but hundreds in a single test. It became known as comprehensive genomic profiling, or comprehensive biomarker testing.

The test that Huff undertook, for instance, looked at over 300 genes in a biopsy of one of the tumours in his neck. It found that his cancer was being driven by an alteration in a gene called *ALK* that is involved in cell growth. Identifying the gene meant he could be treated with a drug called alectinib, which specifically targets this alteration.

Thomas oversaw one of the early trials of ›

comprehensive genomic profiling, launched by the Garvan Institute of Medical Research in Sydney in 2016. It enrolled almost 4000 people with advanced cancers like lung, pancreatic, brain, ovarian, bowel and breast cancer who had exhausted all available treatment options. Samples of their tumours were collected by biopsy and sequenced to examine more than 500 cancer-relevant genes. Over a third of participants were found to have telltale changes in at least one gene that meant they could be recommended a targeted drug. They ended up living almost twice as long as those who weren't matched to a treatment. "It was so successful we had patients flying over from Western Australia so they could take part," says Thomas.

Based on the trial's success, the Australian government agreed to fund a nationwide programme, led by Thomas, which launched in 2023. It offers free comprehensive genomic profiling and tailored treatment recommendations to all Australians with cancers that have low survival prospects. So far, over 26,000 people have signed up.

The outlooks for these participants have become increasingly rosy. "Now, about 48.5 per cent are getting a strong recommendation, meaning we know with reasonable confidence that they will live longer if they receive the recommended treatment," says Thomas.

This is even the case for pancreatic cancer, known as the "worst of the worst" cancer, because it is often survived for only a few months. "We've found that over 30 per cent of patients with pancreatic cancer who receive a good treatment recommendation are alive past five years, compared to 4 per cent who don't get a match," says Thomas. This is mostly because the testing can identify those who are likely to respond to immunotherapy drugs called checkpoint inhibitors, which stop cancer cells from disarming the immune system.

Things are set to get even better, now that a drug called daraxonrasib has just been approved in the US. It targets mutations in a gene called *KRAS* that are commonly found in pancreatic cancer and, in the latest clinical trial, it doubled survival time for those with advanced forms of the cancer.

Another of the most feared malignancies is brain cancer, but precision oncology is starting to make inroads here, too. A new drug called vorasidenib targets mutations in genes called *IDH1* and *IDH2*, which often trigger the growth of brain tumours called gliomas. In a clinical trial, people with gliomas caused by these mutations were 61 per cent less likely to experience worsening of their cancer during the study period if they received vorasidenib compared with a placebo.



THE KENNELLY FAMILY

Jack Kennelly(left) is one of the children who have benefited from the gene-focused approach to cancer treatment being pioneered by those working on the Zero Childhood Cancer initiative in Australia (below left)



CHILDREN'S CANCER INSTITUTE

As the evidence stacks up, countries like Japan, Germany, Norway and Denmark have recently made comprehensive genomic profiling routinely available to all those with advanced cancers as part of their public healthcare systems.

The picture is more complicated in the UK. Most people with advanced cancers are now eligible for comprehensive genomic profiling through National Health Service (NHS) systems across the UK. But despite hundreds of genes being sequenced in each person's tumour as part of a standard test, the results are reported for only around one to 10 genes based on the cancer type and whether treatments are available on the NHS. For example, *ALK* gene alterations are reported for lung cancer because alectinib is already available via the UK's health system.

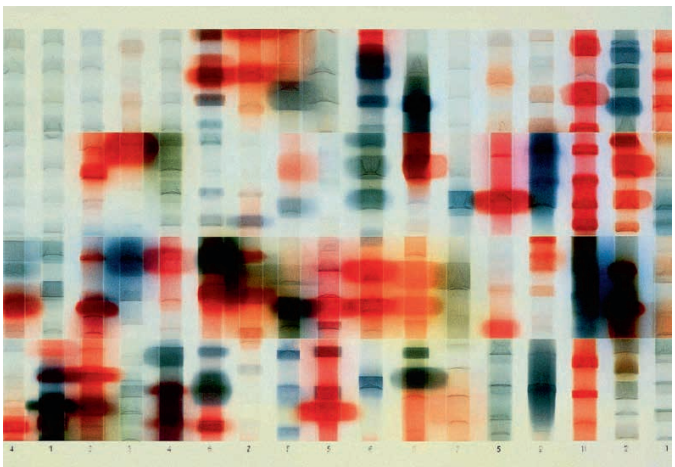
"This supports optimal treatment for standard care, but there's untapped potential for identifying people who could be recruited into clinical trials for new treatments," says Matthew Krebs, an oncologist and researcher at the University of Manchester, UK. To address this, he and his colleagues recently advocated for NHS England to add an extra 30 "research genes" to the small panel currently reported, with the hope of eventually increasing this number.

The US doesn't have a nationally coordinated precision oncology programme, instead relying on a patchwork of services led by various universities and research institutes. "There's quite a bit of competition in terms of different academic centres wanting to have programmes like this for their catchment areas," says Jens Rueter, who runs the Maine Cancer Genomics Initiative at the Jackson Laboratory, an independent research institute.

But despite this, only a minority of people with advanced cancer in the US, about 1 in 4, undergo comprehensive genomic profiling. This has been put down to a lack of awareness and socioeconomic and geographic disparities in access.

As well as improving access to testing, there is a need to improve access to the drugs that are then recommended by that testing. Sometimes, for example, an individual with ovarian cancer is recommended a drug that is currently approved only for breast cancer, meaning they can access it only if they pay for it out of pocket. To address this challenge, some drugs are now being approved as "tumour-agnostic drugs", meaning that they can be prescribed for any cancer, regardless of where it is.

Another headache for precision oncology is the issue of drug resistance. The ability of new drugs to precisely hit their intended targets is both a strength and a weakness – it maximises efficacy and minimises side effects, but also means that, over time, a person's cancer can devise workarounds. That means we have to keep building up our arsenal of targeted drugs.



CAVAN IMAGES/ALAMY

DNA sequencing is revolutionising cancer treatment by pinpointing the exact mutations driving a tumour

Eliminating childhood cancer

In Huff's case, alectinib worked for five years, but then his cancer started progressing again. Fortunately, by that time, a new *ALK*-targeted drug called lorlatinib had been approved. Again, his tumours shrank, and four years on, he says his health is great. He is now married with two children and works in medical technology sales. "Life is really good; I'm just trying to cherish it for as long as I can," he says.

One of the brightest lights in the world of precision oncology is a programme called Zero Childhood Cancer, which is available to all children in Australia with cancer. It goes beyond what is offered in adult programmes in an effort to meet an ambitious goal of cutting childhood cancer deaths down to zero.

When a child is enrolled in the programme, the whole genome of their cancer is sequenced, not just a few hundred genes. Extra tests, called RNA whole transcriptome sequencing and methylation profiling, are performed on top of this to give an even clearer picture of what is driving their cancer. In addition, the genome that the child was born with in their normal cells is sequenced to try to identify why they developed cancer in the first place. Finally, in some cases, the child's cancer cells are grown in a lab so that a robot can test over 100 different cancer drugs on them, or are grown in a mouse to test the potential efficacy of a specific drug.

Jack Kennelly is one of Zero's success stories. He was diagnosed with a fast-growing brain tumour when he was 3 years old, after telling his parents that his head hurt. He was found to have a mutation in a gene called *DICER1*, which had triggered the growth of a rare form of cancer in his brain. *DICER1* normally functions as a tumour-suppressor gene, but certain mutations stop it from working, leading to unchecked cell growth.

No targeted drugs are yet available for *DICER1*-related cancers, but the finding meant that Jack could be given a combination of three chemotherapy drugs that had previously been shown to be most effective for treating his rare form of cancer. This regime, in combination with surgery and radiation, wiped out Jack's brain tumour. "He went in and had an MRI, and they called us and said, 'It's gone'," says his mother, Kristen. "Words can't really express how incredible that feeling was." Jack is now 7, and "just little things like school assemblies are different for us because he's up there doing his thing, he is full of joy, and we're so thankful every day that he is still with us", says Kristen.

The finding that Jack carries a *DICER1* mutation in the genome he was born with also means that his ongoing monitoring can be precisely tailored. Before the age of 8, *DICER1* mutations are most likely to trigger tumour growth in the kidneys, so he currently gets regular scans of this area. "The next thing will be his thyroid," says Kristen. The worry is endless, but the regular surveillance is designed to detect any new tumours while they are still at an early, treatable stage.

Louis Chesler, the director of Children's Cancer Institute in Sydney, which coordinates the Zero programme, says he now believes "more than ever" that children dying from cancer will eventually become a thing of the past. "It's not a matter of if, but when," he says. "It's the most empowering time I can ever think of to be a researcher or clinician in medicine." Emerging technologies like personalised cancer vaccines, blood tests for early detection of cancer recurrences and AI-assisted drug discovery will make precision oncology even more powerful, he says.

For Thomas, precision oncology has become deeply personal. Four years ago, while he was busy setting up Australia's nationwide programme, he himself was diagnosed with cancer and enrolled in his own programme. "Fortunately, I have a cancer that is responding to some of these brilliant treatments," he says. "At this point, I don't think I'm likely to be cured, but I am still able to make a meaningful contribution, both professionally and personally," he adds. "I'm a poster child of what modern medicine can achieve." ▮

"Being able to identify exactly what makes a cancer tick is having an incredible impact"



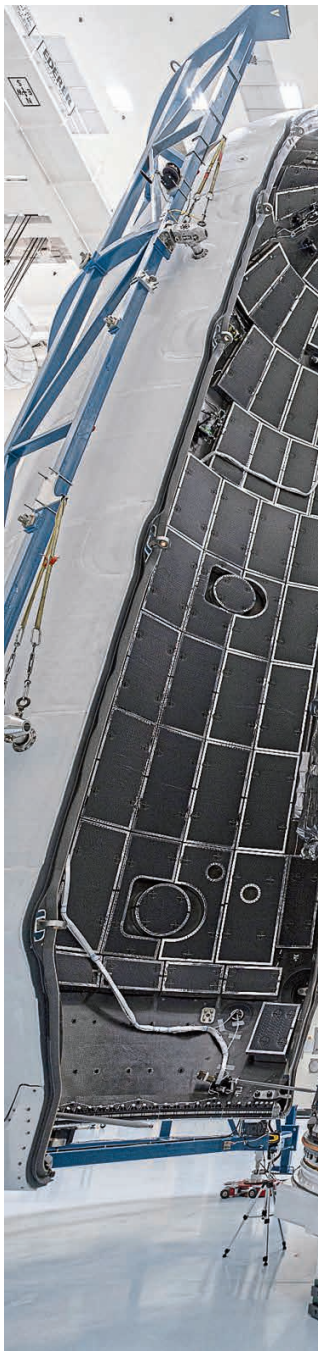
Alice Klein is a reporter at *New Scientist* based in New South Wales, Australia, who mainly writes about health, biology and the environment



Kristen McQuinn is head of the Space Telescope Science Institute's Nancy Grace Roman Space Telescope mission office

"Roman is going to bring us deep insights into the nature of dark energy, but also dark matter"

Once NASA's latest space telescope is fully up and running, it will offer us a panoramic view of the sky. Astrophysicist **Kristen McQuinn** tells Leah Crane how it will go about helping to solve our biggest questions about the cosmos



The Nancy Grace Roman telescope, seen here before launch, is expected to make detailed observations of 12 per cent of the sky

ON 30 August, NASA launched its latest flagship space observatory, the Nancy Grace Roman Space Telescope. It is an incredible piece of technology: it is the same size as the Hubble Space Telescope but with a far broader field of view. Over the course of its five-year primary mission, Roman is expected to observe about 12 per cent of the sky in incredible detail, peering into the distant reaches of the universe. It doesn't sound like a lot, but that is a giant leap from its predecessors – Hubble observed only 0.1 per cent of the sky over 35 years.

This wider view will allow Roman to catch cosmic events as they evolve in real time, and observe an astounding variety of both new and already-known objects in space. But the pièce de résistance will be what it can do for our studies of dark matter and dark energy, the mysterious stuff that makes up about 95 per cent of the universe. Indirect observations tell us that both of these enigmatic components of the cosmos exist, but we know little to nothing about what they actually are or how they work. Roman may be the key that finally unlocks some of their mysteries.

The telescope's powerful 300-megapixel camera was activated two weeks after launch, opening its eyes to the universe as part of its commissioning process. The first photons of





SYDNEY ROHDE/NASA

starlight it took in created fuzzy green rings, an expanse of out-of-focus stars. Kristen McQuinn, a professor at Rutgers University in New Jersey and the head of the Space Telescope Science Institute's Nancy Grace Roman Space Telescope mission office, tells *New Scientist* this is only the beginning. The next steps are to focus the camera, turn on the guidance systems and find out what astonishing things this new observatory will teach us about the universe.

Leah Crane: So, first things first: how was it to finally see that first image?
Kristen McQuinn: It's extremely intense. The teams are on duty 24/7, so there are a lot of people on shifts around the clock. These first images were taken in the middle of the night, but the run-up to this had been going so incredibly well. Leading up to this, we're all kind of on a high.

For the test image, the telescope hasn't been focused yet, and the telescope is also not guiding yet – the telescope is pretty stable when you point it at something, but it can drift a little bit. The test image didn't have the guiding and it was unfocused, and even with that, the image is really spectacular. The lack of focus is really not more than was expected from launch and all of the stressors that come with the launch of a telescope into space: it's telling us that we're in really good shape. The image is so full of promise – everybody is so excited.

Normally, when we get a big new observatory, it's either bigger than the previous ones or it's observing in a unique wavelength of light that we didn't have coverage of before, but Roman isn't either of those things. I don't want to be disrespectful here, but given that, why have we put so much work and investment into it?
No offense taken at all! It's got a mirror similar to Hubble's in size, and it covers some wavelengths that are also overlapping with Hubble and the James Webb Space Telescope, so why do it? The difference with Roman is how much of the sky it can see all at once, its field of view. Hubble and Webb are like a zoom lens on your camera, but Roman is like a panoramic picture. It sees about 100 times larger than the field of Hubble, but it's built in such a way that it can map areas of the sky about 1000 times as fast as Hubble. Every 10 months, we are going to get the equivalent of about 1000 years of Hubble observations.

From a usage standpoint, it's very ›

different. Because Roman can see large areas of the sky, that's what we're going to do. That's why some of the biggest science goals for Roman are understanding dark energy and dark matter, because to probe, both of those you need really, really wide areas of the sky to explore.

How is it that Roman is so much faster than Hubble despite their primary mirrors being the same size?

The actual detector is much larger, and it's positioned in this unique shape. The way Roman is designed, it captures the light in the big primary mirror and it focuses it onto the detector, and because of advances in technology since Hubble was launched in 1990, Roman can bring a larger area into focus. It can also move and map the sky faster than Hubble because of the way that it's built. Its structure is much lighter, less than a quarter of the weight of Hubble. It's also built with a stiffer integrity, and that means that it's got this ability to have rapid torques over short distances. So, you have this stiff, light structure that can pivot and then still itself really, really quickly. You're able to get a bigger field of view, and then you can move the thing between fields of view much, much faster.

Why is it important to get such a huge field of view?

There's this term, cosmic variance, and it says that the universe is, on very large scales, pretty homogeneous. So, if you look at one big area of the universe, it might be similar to all the others, but when you zoom in on any smaller area, everything is different and there are lots of unique things there. So, with Hubble and Webb and other telescopes with smaller fields of view, you can't really know if you're studying unique areas of the universe. Because of this cosmic variance, you need to zoom out to larger areas in the sky to know whether or not what you're seeing is representative of the universe. A bigger field of view allows you to get out of this cosmic variance problem.



SPACEX

After launching on 30 August, the Roman telescope separated from the rocket (below) before reaching orbit

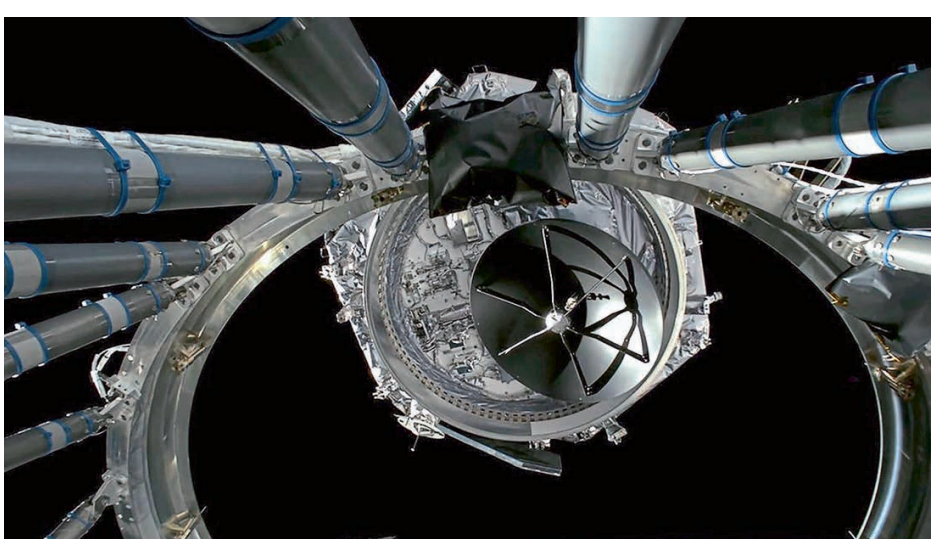
How does that field of view help you with your research?

My own research is on small galaxies – the smaller, the more interesting they are to me, because they're very sensitive to all kinds of dark matter physics and physics of the conditions of the early universe and things that are otherwise hard to probe. With Roman, it's going to be like a galaxy-hunting machine for these small galaxies.

Because these galaxies are so small, they have very shallow gravitational potential wells [meaning that they don't have much mass, and it isn't particularly tightly bound together]. And because they have shallow gravitational potential wells, they are more susceptible to what happened in the early universe, especially during the epoch of reionisation, when copious amounts of UV radiation were produced by massive stars. There's a lot of tie-in between the properties of these galaxies and reionisation physics, and that's tied to dark matter physics because the gravitational potential is set by the amount of dark matter.

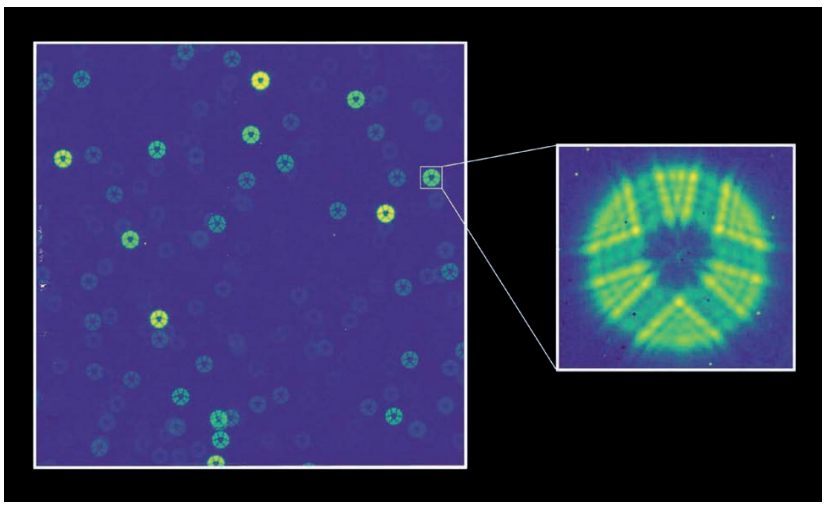
So that's one way Roman tells us about dark matter, but it seems pretty indirect to me. Are there more direct ways, too?

One of the things Roman is going to measure is weak gravitational lensing [where a massive object in the foreground of an image warps the light coming from objects behind it, allowing us to detect and map the relatively nearby mass even if we can't see it directly]. Through that, you are able to measure the clustering and the shape of galaxies being affected by dark matter and dark energy, across the span of time. The weak lensing measurements for Roman are going to be exquisite, because some of the most stringent requirements on the design of the telescope are related



NASA

The first image captured by the Roman telescope was unfocused because of stressors from the launch



NASA'S GODDARD SPACE FLIGHT CENTER, TYLER DESJARDINS (STSCI)

"The first image taken by the telescope is so full of promise – everybody is so excited"

to making this weak lensing measurement. It's a really hard measurement to make, but Roman will be one of the most precisely, if not the most precisely, calibrated telescopes ever.

Do Roman's abilities to measure dark matter tie in at all with what is happening in laboratories and astroparticle observatories right now, as they are investigating the possibility of a dark matter particle detection?
Yes! The clustering of galaxies changes if you have different dark matter particles, or the nature of dark matter is different. And by being able to measure this so exquisitely with Roman, it can help constrain or narrow down some of the dark matter candidates. If you've got a lighter, warmer dark matter particle, then the clustering of galaxies is not quite so compact, and if you have a colder or heavier particle, then the clustering of galaxies is slightly more compact. But it's a very subtle effect, so that's why you need to build a telescope like Roman to do it.

As exciting as that is, dark energy is really more of the main target for Roman, right? And that's a little more complicated than dark matter, because it's something that is making the whole universe expand, not a substance that is localised in one place. So aside from the lensing, how are we going to study that?
One way we measure the expansion of the universe is through type Ia supernovae explosions. These are very predictable in the way the light from their explosions changes with time, the brightness and their characteristics, which is why we call them "standard candles". And so, because of that, if you see one go off and you take the right observations, you can figure out how far away it is. Couple that with the velocity of the galaxy hosting the supernova explosion, and you can really start mapping out how the universe is moving over large scales.

But the capabilities that we have today can really measure supernovae out only to a certain distance, and after that, we don't really have those capabilities. But Roman brings those capabilities, so now we can actually measure type of supernova, out through very far cosmic distances, and have one single method with one type of observatory without a lot of systematic uncertainties to be able to measure the expansion rate of the universe. That can help us understand the acceleration and whether it has changed at different epochs of time.

The other way it does that is by measuring baryonic acoustic oscillations, which are these sound waves that got frozen into the universe at very early times. These are more of a standard ruler technique [because we know what size they ought to be, so by measuring the size that they appear, we can tell their distance]. And Roman will be able to study the acceleration in dark energy based on that as well.

The big question, I think, is: will Roman finally answer the question of what dark matter and dark energy really are, or is it just one more step on a long road towards the answer?
I think Roman is going to bring us new insights into these big questions. I sincerely do. It's difficult to predict exactly the level of breakthrough, but we've never had anything like Roman before. There are other surveys, and we're going to learn important things from all of them in different ways. Roman isn't the only game in town.

But it's the newest game in town.
It is the newest game in town. And I'm a little biased, but it's the best game in town. I think Roman is going to bring us deep insights into the nature of particularly dark energy, but also dark matter, and I think we're going to learn a lot. Will it completely unravel all of the mysteries associated with these phenomena? Unlikely. You know, the universe is a very complex and amazing place, and the more we learn, the more we know there's more to learn. But I think Roman isn't just one step on a long road. We are going to get some breakthrough discoveries with the Roman observations. ▮

This is an edited extract from our subscriber newsletter Lost in Space-Time. Sign up at newscientist.com/sign-up/lost-in-space-time



Leah Crane is a senior reporter at *New Scientist*, covering topics ranging from private space flight and cosmology to quantum mechanics

EARLY BIRD CHRISTMAS OFFER

What on Earth! MAGAZINE

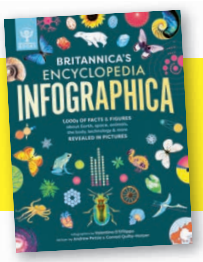
Open a world of discovery!

SAVE 60%

- ✓ A brilliant monthly magazine for ages 7–14
- ✓ Amazing facts, photos, and screen-free fun
- ✓ All their favourite topics, from animals, space and nature to art, sport, science and inventions
- ✓ Puzzles, quizzes, games, activities and jokes
- ✓ 10 major awards, including Children's Magazine of the Year
- ✓ Help them fall in love with reading and learning



OFFER ENDS 31 OCT



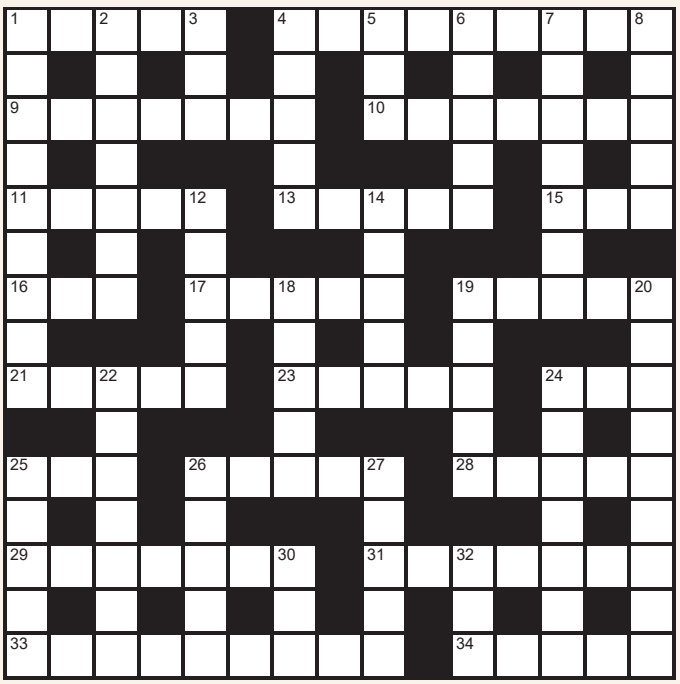
INCLUDES BRITANNICA KIDS' ENCYCLOPEDIA WORTH £27*

*Free book for new 12-month subscribers only. While stocks last. whatonearthmag.com/terms

whatonearthmag.com/ns

0330 333 0261

Quick crossword #219 *Set by Richard Smyth*



Scribble zone

Cryptic crossword next week

ACROSS

1 Living things (5)
4 Winged insect such as a red admiral (9)
9 Openings for the entry of fluids or gases (7)
10 Dream-like or relating to dreams (7)
11 Distance from top to bottom or front to back (5)
13 Venomous snake (5)
15 Age (3)
16 Bodily fluid (3)
17 Type of parrot (5)
19 H_2O (5)
21 Genus or clade, say (5)
23 State of matter (5)
24 τ (3)
25 For each (3)
26 Film containing micro-images (5)
28 Bone in the neck (5)
29 Not preserved or backed up (7)
31 Electrical device also known as a selsyn (7)
33 Female hormone (9)
34 ___'s razor, principle in problem solving (5)

DOWN

1 Zone of limited visibility (5,4)
2 Eight-limbed mollusc (7)
3 Puffin or guillemot, say (3)
4 General-purpose programming language invented in 1964 (5)
5 Cube root of 8 (3)
6 Bowel treatment (5)
7 Undergo a type of anaerobic metabolism (7)
8 Spiny plant related to the agave (5)
12 Layer of mucosal tissue in the vulva (5)
14 Gut (5)
18 Reciprocal trigonometrical function (abbreviated) (5)
19 Distance from one side to another (5)
20 Nematode (9)
22 Dryness (of skin) (7)
24 ___ cascade, series of linked ecological interactions (7)
25 Demoted planet (5)
26 High temperature (5)
27 Fluorescent acidic compound (5)
30 Canine (3)
32 Prefix meaning "new" (3)

Quick quiz #374
set by Chelsea Whyte

1 A semi-aquatic mammal that lived alongside the dinosaurs was found to be able to regrow which body part throughout its life?

2 What defensive tool is being used to thwart fibre-guided attack drones in Ukraine?

3 Machine learning used to analyse elephants' calls found that these animals do what when they make noise together?

4 A small hoard of coins found near the Vatican suggests that a certain segment of the population held privileged status in early-medieval Rome. Who were they?

5 Cutting out which food from your diet could protect your gut microbiome while taking a course of antibiotics?

Block Universe
created by Puzzlr

The goal is to divide the grid into non-overlapping rectangles. Each rectangle must contain exactly one number, and have an area equal to that number. For example, a 6 should have a rectangle covering 6 squares.

						4		3
			16					
				12			10	
	6		3					7
6					2			
			2					
	6						4	
		3	2	4	2		2	
					5			

Answers to this week's puzzles on page 47

Our games are now playable online
newscientist.com/games

Don't cry

Are humans the only animals that cry? If so, why?

Mike Follows
Sutton Coldfield, West Midlands, UK
It depends what we mean by "cry". If we mean making a sound when distressed, humans are certainly not alone. Distress calls are widespread in the animal kingdom and are examples of an ancient form of communication. A hungry, frightened or separated youngster may attract food or protection by calling, while adults may warn others of danger.

Humans are unusual because we also cry tears. Lots of animals produce tears, but ordinarily these have nothing to do with being sad. Basal tears keep the eye moist, while reflex tears respond to irritants such as smoke or onions. Emotional tears, by contrast, can accompany sadness, grief, anger, relief, joy and even laughter.

The most plausible explanation for emotional tears is their use in communication. They provide a conspicuous signal of distress or vulnerability and may encourage sympathy or reduce aggression. But if tears are such an effective signal, why don't newborn babies use them from the outset?

> **"Emotional tears may not originally have evolved as a signal, but simply accompanied intense emotional arousal"**

Babies certainly cry, but initially sound is the important signal because a baby needs to attract a parent who may not be looking at them. A loud cry works at a distance and in darkness; tears only work if somebody can see your face.

Emotional tears may not originally have evolved as a signal, but simply accompanied intense emotional arousal. Once other humans began responding to a tearful face, however, natural selection could reinforce its signalling function. But any useful signal is open to exploitation. Someone able to cry convincingly without experiencing the corresponding emotion may manipulate the response of others, known as crocodile tears.

Perhaps this is why tears alone aren't enough to convince us that someone is genuinely upset. We also look for confirmation in facial expression, voice, circumstances and previous behaviour. Actors complicate matters further because some can deliberately summon a genuine emotional response and cry on demand.

The expression crocodile tears comes from the old belief that crocodiles wept while eating their victims, apparently regretting an act while continuing to benefit from it. The idea is old enough for William Shakespeare to have used it in *Othello*. Curiously, there is some real biology behind the story. Crocodilians (which include alligators, caimans and gharials, as well as crocodiles) have lacrimal glands and can appear to shed tears while feeding. The secretions have nothing to do with emotion.



HEMIS/ALAMY

This week's new questions

Colour conundrum **Why do humans enjoy black-and-white photos or videos, but have less tolerance for images that have only two other colours?** ***Justin Butler****, Melbourne, Australia*

Paired up **If two chromosomes are better than one, why not three? Wouldn't this give you a better chance of surviving environmental change?** ***Jim Wegryn****, Dimondale, Michigan, US*

Why do we like black-and-white images, but not images of just two other colours?

Jog on

Is it faster to jog up the stairs or take them two at a time?

Andrew Hardwick
Ipswich, Suffolk, UK
Two at a time, definitely. I say this with personal experience from years in an office 135 stairs up.

Humans have evolved to scramble up brief steep slopes with long strides. Jogging was created to improve fitness, not speed.

Use the handrail. Not just for adding pull on the way up, but for maintaining momentum on turns in the staircase.

As for down? I could do only one at a time, but a friend with longer legs could do two at a time on the way down, too. I could jump the last three steps to each turn landing, though.

David Cordiner
Birmingham, UK
I used to run up stairs two or three at a time (I couldn't do it now). It is certainly faster than using every step. Just walking two at a time is slow.

Hear that?

Why don't dogs look up when a low-flying plane or helicopter passes overhead? I've never seen one do this. (continued)

Wayne Lewis
Lubbock, Texas, US
While this is unusual, there are some dogs that will look at moving objects in the sky.

We had a male collie that regularly did this. People would comment about it, since it is rare for dogs.

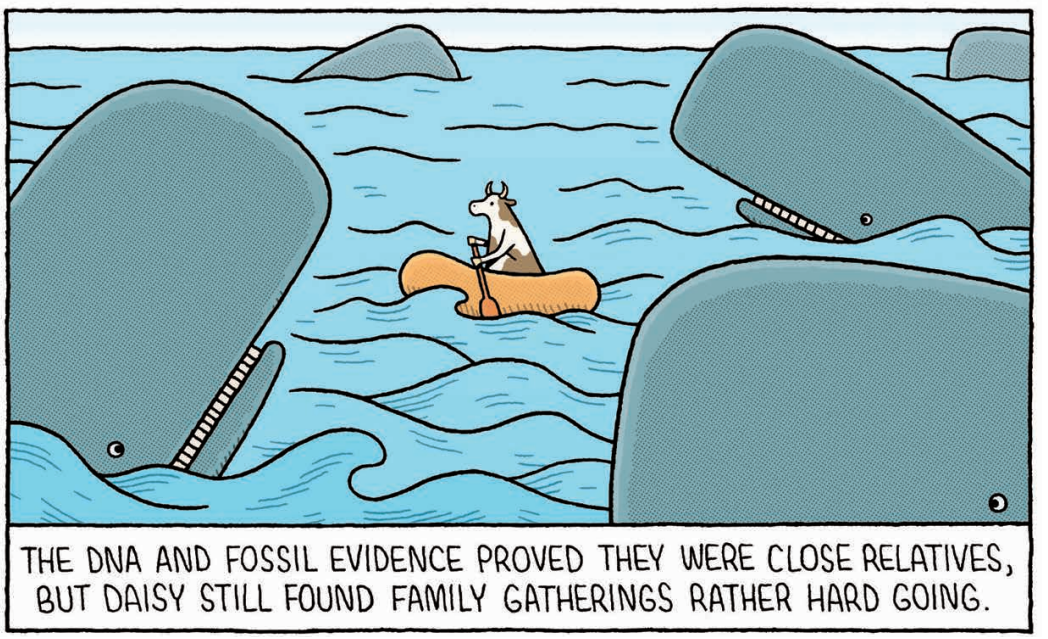
Helicopters are usually clearly heard first and then located, at least by humans. My collie would sometimes detect them before I did. He would raise his head and follow with his eyes. I don't clearly remember any motion of his ears.

This was before the local air force base was closed. Planes included military jets, passenger jets and private planes. My collie would recognise and follow each type. He would do this more frequently if they were close enough to have a significant profile or be clearly audible, but would sometimes follow them when neither condition was met.

On rare occasions, there would

Want to send us a question or answer?
Email us at lastword@newscientist.com
Questions should be about everyday science phenomena
Full terms and conditions at newscientist.com/lw-terms

Tom Gauld
for New Scientist



be hot air balloons, clearly visible but silent. The collie seemed to watch them more closely and act puzzled.

Living in a wintering area for Canadian geese, all of my collies would watch them and want to chase them when flocks were taking off. The geese would frequently be honking a lot. Only one of my collies would pay any attention to them when a flock was passing overhead or coming in to land.

My collie didn't make any sounds when observing such, except for especially low hot air balloons. But he was definitely following them with his raised head and eyes.

Keith Wright
Manchester, UK

Further to Ian Tindall's answer (12 September), I agree that dogs do look up within buildings, even if they don't at planes outside.

On a hot day, I drove around 290 kilometres with my dog in the back of my hatchback car. On arrival at my brother's flat, which she had never visited before, she went straight to the kitchen (presumably by smell) and did the classic begging pose next to the sink. She clearly knew exactly which part of the worktops would deliver water, despite being too small to see the taps from the ground. Yes, this was by smell, but I think it was also by the appearance of the sink edge, and her familiarity with ours at home.

"Perhaps many dogs don't look at aeroplanes because we don't, and that if we do, it results in nothing they want"

As I walked towards the sink with her water bowl in hand, she was very definitely looking at me and the edge of the sink alternately, as well as enthusiastically wagging her tail. She not only knew where the water was in a strange flat, but knew I would understand her directional looks.

Perhaps many dogs (including that one) don't look at aeroplanes because we don't, and that if we do, it results in nothing that meets their needs.

Alan Harding
London, UK

Further to the answers already published, I wonder if there is another explanation.

Dogs have a very strong sense of smell, and many breeds don't have particularly good eyesight, so live in an olfactory world rather than a visual one.

Passing planes and helicopters wouldn't register with dogs living in a world of smells. This might also explain Dyane Silvester's Alsatian's ability to locate her in an upstairs room (8 August) – no visual or geometrical ability was required, just following a scent trail.

As a child, my sister and I used to play hide and seek in the local woods with our pet corgi. She never had the slightest difficulty finding our hiding places, and clearly regarded the entire game as rather silly. ▮

Answers

Quick crossword #219

ACROSS 1 Biota, **4** Butterfly, **9** Intakes, **10** Oneiric, **11** Depth, **13** Cobra, **15** Era, **16** Pus, **17** Macaw, **19** Water, **21** Taxon, **23** Solid, **24** Tau, **25** Per, **26** Fiche, **28** Hyoid, **29** Unsaved, **31** Synchro, **33** Oestrogen, **34** Occam

DOWN 1 Blind spot, **2** Octopus, **3** Auk, **4** BASIC, **5** Two, **6** Enema, **7** Ferment, **8** Yucca, **12** Hymen, **14** Bowel, **18** Cosec, **19** Width, **20** Roundworm, **22** Xerosis, **24** Trophic, **25** Pluto, **26** Fever, **27** Eosin, **30** Dog, **32** Neo

Quick quiz #374

1 Teeth

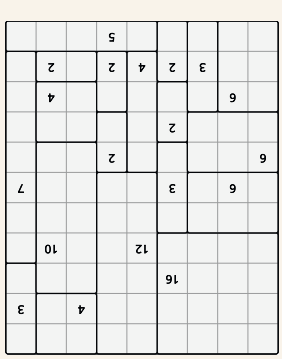
2 Motorised barbed fences

3 Harmonise

4 Anglo-Saxons

5 Sugar

Block Universe *Solution*



To boldly cook

Various Trekkies in the *New Scientist* office have pointed out to Feedback that this year is the 60th anniversary of the venerable sci-fi franchise *Star Trek* (TV columnist Bethan Ackerley even wrote a whole article about it). Naturally, its corporate owners have marked the occasion by cancelling all the television shows still in production. A movie is currently in development, but since that has been continuously true ever since the last cinematic release in, er, 2016, it seems best not to hold one's breath.

In the absence of, say, a special episode where the characters in a current show are inserted *Forrest Gump*-style into an episode of the original programme, fans must make do with what they have, and what they have is apparently an expensive range of *Star Trek*-branded cookware, one item of which was briefly sent into space.

For some reason, *Star Trek* has partnered with Le Creuset, makers of distinctively orange cookware. One can now buy a selection of cooking items, including various casserole dishes emblazoned with trekanalia, a mug with a handle shaped like one of Spock's pointed ears, and a spoon rest in the form of a Vulcan salute. A small prize, also, to whoever came up with the idea of the communi-grater.

To draw attention to this new range, Le Creuset hired a company called Sent Into Space, which describe themselves as "the industry leaders in filming products, items, and bespoke activations in space". Sent Into Space used a balloon to take a *Star Trek* casserole dish to an altitude of 35 kilometres, which isn't a sentence Feedback ever expected to write. One can, if one so chooses, watch this event on the Le Creuset YouTube channel, which is a thing that exists.

Chief subeditor Kelsey Hayes learned of this after being sent a promotional email about it. She notes that 35 km is "well below the Kármán line", which is one standard definition for the edge of space and is somewhat arbitrarily set at 100 km altitude. So, it's a matter of debate whether the cookpot went to space or just really high.

Feedback will say this: we prefer this stunt to that time Elon Musk sent a Tesla Roadster into space. That car is now orbiting the sun and will litter the solar system forever. In contrast, the Le Creuset cocotte was parachuted back to Earth.

Happily, the landing took place in an isolated field. As Kelsey says: "Imagine getting beaned by a Le Creuset Dutch oven dropped from 110,000 feet."

Twisteddoodles *for New Scientist*



Got a story for Feedback?
Send it to feedback@newscientist.com
or New Scientist, 9 Derry Street, London, W8 5HY
Consideration of items sent in the post will be delayed

Hey good-looking

Feedback is always keen on a new scientific acronym, and this one is new to us: self-assessed physical attractiveness (SAPA). As the name suggests, it refers to a person's opinion of their own hotness. And according to a recent study, a person's SAPA can be linked to a host of other traits.

In the journal *Evolutionary Psychology*, two researchers report asking hundreds of volunteers to rate their own attractiveness and then answer questions to see what correlated with SAPA. "Participants rated their attractiveness on a 0–100 scale," which Feedback finds immediately baffling because how do you choose between, say, 76 and 77? Surely 1 to 10 would do.

This one was sent our way by assistant news editor Alexandra Thompson, who flagged a key line in the abstract – "higher self-rated attractiveness was associated with being male" – and simply said "LOL". We must tell you that higher SAPA was also associated with holding a university degree, greater optimism, being more religious, having a liberal political orientation and being able to tolerate ambiguity. In contrast, "cognitive ability showed a small negative association with SAPA".

It seems to follow that the people with the highest opinion of their own attractiveness are men who have university degrees despite lacking cognitive ability. Upon having this realisation, Feedback had a flashback to our days as an undergraduate, and we can see the truth in it.

The researchers also did personality tests, but since they used the DISC test invented by the creator of Wonder Woman, which is decidedly non-evidence-based, we won't bother about the results.

The point of all this is to try to understand the dynamics of our society. The authors write: "There is a vast literature on the beauty premium, which suggests that, in males, and possibly more in females, physical attractiveness is a considerable asset in all competitive situations such as mate selection."

Feedback cannot quite get over how many people think of dating as inherently competitive. Our experience is that the main obstacle to securing a healthy adult relationship isn't competitors who are more attractive, but rather one's own devastating personality flaws.

Maybe that's because Feedback is so ridiculously good-looking.

Monks gone wild

Our mission to explore the world of unfortunate tech-induced abbreviations and truncations is ongoing. Duncan King tells us that, in the 1970s, his job included teaching students how to use the Science Citation Index, an index of thousands of leading journals.

Duncan explains that the index used a KeyWord In Context system. However, on occasion this led to peculiar side effects. Notably, an entry that was supposed to read as "Atherosclerotic tendencies in monks" would be split by the format, appearing instead as "ATHEROSCL EROTIC TENDENCIES IN MONKS".

New Scientist Discovery Tours

Uncover Indonesia in the footsteps of Alfred Russel Wallace







Alfred Russel Wallace expedition cruise: Indonesia

23 January 2027 & 23 January 2028 | 13 days

Explore Indonesia's Maluku and Raja Ampat islands, where extraordinary biodiversity shaped Alfred Russel Wallace's pioneering work on evolution. Accompanied by Wallace expert George Beccaloni, encounter remarkable wildlife, from birds and butterflies to the rich marine life of one of the world's most biodiverse regions.

- **Travel aboard the beautifully crafted, 42-metre Ombak Putih, an exclusive 22-berth private yacht**
- **Explore by kayak, paddleboard, snorkel and sunrise guided walks, discovering unique species as you go**
- **Visit key sites associated with Wallace's research and explore the evolutionary forces behind Indonesia's extraordinary biodiversity**



Accompanied by entomologist and evolutionary biologist, Dr George Beccaloni

George has studied the life and work of Wallace for over 20 years and will share his discoveries and ideas through lectures and walking seminars at key sites.

Science-led journeys for curious minds

newscientist.com/tours tours@newscientist.com @newscientistdiscovery

SCAN TO FIND OUT MORE

omegawatches.com



Speedmaster

Iconic since 1957

SPEEDMASTER MOONWATCH PROFESSIONAL
Co-Axial Master Chronometer

OMEGA